

A Preference for Traceable Agency

An Axiomatization of Expected Utility plus Mutual Information*

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Abstract

An action can matter twice: through the payoff it produces and the visible trace it leaves. A channel maps actions into consequences; preferences rank action lotteries in each channel. Data-processing conditions and a recordwise sure-thing principle, with standard axioms, characterize expected utility plus λ times the mutual information between action and consequence. The theorem identifies λ , the price of trace in payoff units; expected utility is the case $\lambda = 0$. The criterion reads the coupling of act and consequence, not consequences alone; randomisation can be strictly optimal. When each action leaves a distinct trace, the optimal lottery has multinomial-logit probabilities; when traces overlap, substitution depends on the similarity of their consequence distributions and independence of irrelevant alternatives can fail.

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1 Introduction

Shower upon him every earthly blessing ... He would even risk his cakes and would deliberately desire the most fatal rubbish, the most uneconomical absurdity ... simply in order to prove to himself—as though that were so necessary—that men still are men and not the keys of a piano. ... If you say that all this, too, can be calculated and tabulated ... then man would purposely go mad in order to be rid of reason and gain his point!

— Fyodor Dostoevsky, *Notes from the Underground*, Part I, Chapter VIII [20]

Dostoevsky’s *Underground Man* faces a choice problem. The cakes are material payoff, and he will risk them so that his conduct can attest to being his own, not the mechanical implication of a table. Yet the proof he seeks is self-defeating. An act fixed in advance cannot provide it, and deliberate rebellion, too, can be “calculated and tabulated.” So he would “purposely go mad in order to be rid of reason”—not because madness is what he values, but because it is his last imagined way of preventing the table from closing before he acts. This paper takes the evidentiary idea literally. In Dostoevsky’s example, conduct is observed directly: the act supplies its own evidence. We consider the general case, in which an act is observed only through a possibly noisy payoff-bearing outcome and an explicit visible record.

For visible consequences to carry information about an act, two conditions are needed. There must be uncertainty beforehand about which act will occur, and the consequence must discriminate among the possible acts. Either condition can fail on its own. A lottery over acts leaves no trace when every act generates the same distribution of visible consequences; a consequence that perfectly reveals the act conveys nothing new when the act was already known. We call the intrinsic preference for the two together a preference for *traceable agency*. The paper axiomatizes it jointly with ordinary preferences over payoff risk.

Let A be a finite set of actions, O a finite set of payoff-bearing outcomes, R a finite set of explicit visible records, and $K : A \rightarrow \Delta(O \times R)$ the channel relating actions to visible consequences. An action lottery $q \in \Delta(A)$ gives the distribution of the realised action. The primitive is a family $\{\succeq_K\}_K$, with one binary relation $\succeq_K$ on $\Delta(A)$ for each fixed channel K . The statement $q \succeq_K p$ means that q is weakly preferred to p in environment K . Eight axioms are imposed on this family. To compare lotteries governed by different channels, the channels are placed side by side as blocks of one larger environment; duplicating a channel or adding an unused block does not alter the comparison at issue. Two processing axioms say that the agent never gains when her record is garbled or her action variable is processed noisily, and a benchmark comparison rules out indifference to trace. A recordwise sure-thing principle for records that arrive in stages supplies the cardinal structure. No audience appears in the model, and the taste is content-neutral: it prices how much the consequence reveals, not what it reveals.

The main result is an if-and-only-if characterization. A family $\{\succeq_K\}_K$ satisfies the eight axioms if and only if there are a nonconstant payoff utility $u : O \rightarrow \mathbb{R}$ and $\lambda > 0$ such that, for every finite joint channel K and every $q, p \in \Delta(A)$,

$$q \succeq_K p \iff \mathbb{E}_{qK}[u(O)] + \lambda I_{qK}(A; O, R) \geq \mathbb{E}_{pK}[u(O)] + \lambda I_{pK}(A; O, R),$$

where $I_{qK}(A; O, R)$ is the mutual information between the action and its visible consequence. We call such families *trace-tempered*. The utility u is unique up to positive affine transformations, λ is measured in payoff units per unit of information, and the same pair prices comparisons across channels, embedded as blocks of one larger environment. A sign trichotomy completes the picture: reversing the axioms that carry the taste for trace characterizes $\lambda < 0$ —a preference for deniability—while weakening them to indifference yields expected utility, $\lambda = 0$.

The information term has a simple interpretation. Observing (O, R) induces a posterior belief about the realised action; traceability is the expected reduction in that uncertainty, measured

by Shannon entropy H . If visible consequences are independent of actions, the value is zero. If the action is known in advance because q is degenerate, the value is also zero. If each action in the support of q produces a distinct visible consequence, the value is $H(q)$ —Dostoevsky’s case, in which conduct is observed directly.¹ Thus the criterion is not a taste for randomisation as such. Randomisation matters only because it creates uncertainty about agency for the visible consequence to resolve.

Traceable agency is thus a taste for the coupling of act and consequence, not for consequences alone. A four-action channel turns this into a direct test. Every action pays the same amount and writes one of two symbols on the record. Action a_1 always writes the first symbol and a_2 the second; a_3 and a_4 each write either symbol with equal probability. The choice is between a fair coin over $\{a_1, a_2\}$ and a fair coin over $\{a_3, a_4\}$. The two coins randomise identically and induce the same joint distribution of payoff and record. Under the first coin the record identifies the realised action; under the second it reveals nothing. Any criterion that reads only the distribution of consequences is indifferent between the coins; so is a taste for randomisation as such. Traceable agency predicts a strict preference for the first. Because the record affects neither payoffs nor any later decision, the preference cannot be instrumental.² The comparison identifies a value of act–consequence dependence; by itself it does not distinguish trace from control or identify the psychological source of that value.³

This comparison has not been run. The nearest evidence comes from two experiments. Bartling, Fehr, and Herz elicit each principal’s point of indifference between retaining and delegating the right to choose a project and the effort devoted to it. At that point, the delegation lottery has a certainty equivalent 16.7% higher on average than the lottery produced by the principal’s own choices, when both are valued outside the delegation setting [6]. This is consistent with a preference for trace, since retaining control makes the principal’s choices determine the payoff lottery, but it may instead reflect autonomy or some other value of deciding for oneself. Mistry and Liljeholm, and later Norton and Liljeholm, ask subjects to select a room containing two slot machines and then gamble in that room. Subjects are more likely to choose a high-divergence room over a zero-divergence room when they choose between its machines themselves than when a computer chooses for them, holding current expected payoffs and outcome entropies fixed [36, 37]. With equal weight on the two machines, the divergence between their token distributions is the mutual information between the machine and the token. This also admits a trace interpretation: in self-play, the token is evidence about a voluntarily selected machine. But token values could change after the room was selected, allowing subjects in self-play to adapt their machine choices. Thus trace is confounded with the value of deciding in the first design and with that of adapting in the second. The test proposed above removes both confounds.

Section 2 sets out the domain and the eight axioms. Section 3 states the representation theorem, with a proof sketch, and derives uniqueness, the sign trichotomy, and the independence of the axioms. Section 4 presents a two-treatment test that can reject the model; derives deliberate randomisation, with multinomial logit as the boundary case when every action leaves its own mark; prices garblings of the record; measures λ from choices; and characterises when one agent is more trace-seeking than another. Section 5 reviews the literature, in particular rational inattention, where mutual information is a cost of attention rather than an object of taste, and the axiomatic characterisations of information indices. Section 6 discusses interpretation and scope. The proofs are in the appendices: the pure-trace characterisation in Appendix A, the completion of Theorem 1 in Appendix B, and the remaining results in Appendix C.

¹If the visible-consequence supports of all channel rows are pairwise disjoint, the unique optimal lottery has multinomial-logit weights. Overlap can make independence of irrelevant alternatives fail; see Remark 7.

²A second treatment completes the design: repeat the choice in a channel that writes a fair record independently of the realised action. Every representation then assigns both flips the same value, whatever u and λ , so a strict preference there rejects the model. Section 4 develops the test and what it excludes.

³The information term admits a control interpretation. Read backward: it measures trace. Read forward, it measures how strongly the action controls the distribution of visible consequences. We return to this in Section 6.

2 The model

Fix a finite set O of *payoff-bearing outcomes*, with $|O| \geq 2$. Let A be a nonempty finite set of *actions* and R a nonempty finite set of *explicit visible records*. A joint channel is a stochastic kernel

$$K : A \longrightarrow \Delta(O \times R).$$

For each finite joint channel K , the primitive behavioural datum is a binary relation

$$\succeq_K \quad \text{on} \quad \Delta(A).$$

The interpretation is that q is a lottery over actions used by the agent and $q \succeq_K p$ means that, in the fixed environment K , q is evaluated at least as highly as p . Within that environment, the lottery is the procedure being evaluated and the channel is fixed.⁴

Axioms are imposed on the family $\{\succeq_K\}_K$. The first two axioms are standard, and Axiom (A3) is the usual nondegeneracy condition on material outcomes. Axioms (A4), (A6), and (A7) carry the preference for trace. Axiom (A5) puts the separate within-channel rankings on a common scale, while Axiom (A8) imposes consistency across stages; neither fixes the sign of the trace term.

(A1) Weak order. For every finite joint channel K , $\succeq_K$ is complete and transitive on $\Delta(A)$.

(A2) Continuity. For each fixed triple of alphabets (A, O, R) , the set

$$\{(K, q, p) : q \succeq_K p\} \subseteq \Delta(O \times R)^A \times \Delta(A)^2$$

is closed.

(A3) Material relevance. There are distinct outcomes $o^+, o^- \in O$ and a two-action channel K with a one-point record set in which a^+ delivers o^+ for sure and a^- delivers o^- for sure. For this channel,

$$\delta_{a^+} \succ_K \delta_{a^-}.$$

The agent strictly ranks at least two outcomes when each is delivered for sure and the two actions leave the same record.

(A4) Trace relevance. There is an outcome $o^* \in O$ and a four-action channel K in which every action delivers o^* for sure. Actions a_1 and a_2 leave distinct records r_1 and r_2 , respectively, while each of a_3 and a_4 leaves either record with equal probability. For this channel,

$$\frac{1}{2}(\delta_{a_1} + \delta_{a_2}) \succ_K \frac{1}{2}(\delta_{a_3} + \delta_{a_4}).$$

Both are fair lotteries and produce the same distribution of visible consequences. The axiom requires a strict preference for the first, whose record identifies the realised action, over the second, whose record does not.

⁴Why not a single preference over lotteries on $A \times O \times R$? Every joint law is $q \otimes K$ for some pair, so the domain is rich enough. But in any one choice situation the channel is fixed: the agent moves the marginal on A , never the coupling. A preference over all joint laws would rank couplings, and no choice reveals those rankings.

Comparison environments. Some axioms compare lotteries carried by different channels. Each relation $\succeq_K$ ranks lotteries within one channel, so such comparisons need a domain in which two channels appear in a single preference statement. A larger channel that contains both as labelled blocks provides it.

For $K : A \rightarrow \Delta(O \times R)$ and $L : B \rightarrow \Delta(O \times R_L)$, let $K \sqcup L$ be the block-diagonal joint channel—the *block environment*—with action set $(A \times \{0\}) \cup (B \times \{1\})$, payoff set O , and record set $(R \times \{0\}) \cup (R_L \times \{1\})$: each block operates its own channel, and all cross-block probabilities are zero. For $q \in \Delta(A)$ and $p \in \Delta(B)$, let q^0 and p^1 denote their block-supported copies, defined on the tagged action set by

$$\begin{aligned} q^0(a, 0) &:= q(a), & q^0(b, 1) &:= 0, \\ p^1(a, 0) &:= 0, & p^1(b, 1) &:= p(b), \end{aligned} \quad (a \in A, b \in B).$$

Under q^0 , the visible consequence is generated by K after the action is drawn according to q ; under p^1 , it is generated by L after the action is drawn according to p . So a choice between them is also a choice between the two channels, made inside a single environment. Write

$$(q, K) \succeq (p, L) \iff q^0 \succeq_{K \sqcup L} p^1.$$

Each instance of this shorthand is a single comparison inside the two-block environment $K \sqcup L$, which is built from the two channels; $\succ$ and $\sim$ denote the asymmetric and symmetric parts of this shorthand relation. The pair (q, K) has no meaning in isolation.⁵ The construction extends to any finite family $\{K_i : A_i \rightarrow \Delta(O \times R_i)\}_{i \in I}$: write $\bigsqcup_{i \in I} K_i$ for the analogous block environment and q_i^i for the lottery that assigns $q_i(a)$ to (a, i) and zero to every other block.

(A5) Block-comparison coherence. *Duplication:* for every $K : A \rightarrow \Delta(O \times R)$ and all $q, p \in \Delta(A)$,

$$q \succeq_K p \iff (q, K) \succeq (p, K).$$

Irrelevant blocks: for every finite block environment $\bigsqcup_{k \in I} K_k$, every ordered pair of distinct blocks $i, j \in I$, and all $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in I} K_k} q_j^j \iff (q_i, K_i) \succeq (q_j, K_j).$$

Comparing q and p across two tagged copies of K is just comparing them in K ; adding blocks not involved in a comparison leaves it unchanged.

Processing. The next two axioms use two ways of transforming an environment. A *record processor* is a stochastic kernel $T : O \times R \rightarrow \Delta(O \times R')$ that garbles the record without changing the payoff: $T(o', r' \mid o, r) = 0$ whenever $o' \neq o$. The garbled channel is simply the matrix product $KT : A \rightarrow \Delta(O \times R')$,

$$(KT)(o', r' \mid a) := \sum_{o, r} K(o, r \mid a) T(o', r' \mid o, r).$$

An *action processor* is a stochastic kernel $S : A \rightarrow \Delta(B)$ that replaces the action variable A by a processed action variable B . Given $q \in \Delta(A)$, it induces the lottery $qS \in \Delta(B)$,

$$(qS)(b) := \sum_a q(a) S(b \mid a).$$

⁵Formally, $(K \sqcup L)(o, (r, 0) \mid (a, 0)) = K(o, r \mid a)$ and $(K \sqcup L)(o, (r_L, 1) \mid (b, 1)) = L(o, r_L \mid b)$, with every other entry zero. In the binary case, with $A = \{a_1, a_2\}$ and $B = \{b_1, b_2\}$, the block environment has the four tagged actions $(a_1, 0)$, $(a_2, 0)$, $(b_1, 1)$, $(b_2, 1)$; a lottery $q = (q(a_1), q(a_2))$ becomes $q^0 = (q(a_1), q(a_2); 0, 0)$ and $p = (p(b_1), p(b_2))$ becomes $p^1 = (0, 0; p(b_1), p(b_2))$, the semicolon separating the blocks. Then $(q, K) \succeq (p, L)$ abbreviates $q^0 \succeq_{K \sqcup L} p^1$: one ordinary comparison of two lotteries in this one four-action environment.

A *Bayesian pushforward* of K by S at q is any channel $\widehat{K} : B \rightarrow \Delta(O \times R)$ satisfying⁶

$$(qS)(b) \widehat{K}(o, r \mid b) = \sum_a q(a) S(b \mid a) K(o, r \mid a) \quad \text{for every } (b, o, r). \quad (1)$$

In the processed environment, the alternatives are the noisy names b , while the visible consequences remain (O, R) . Choosing qS under $\widehat{K}$ replaces dependence on the original action with dependence on its noisy name, without changing the marginal distribution of visible consequences.

(A6) Record data processing. For every $K : A \rightarrow \Delta(O \times R)$, every record processor T , and every $q \in \Delta(A)$,

$$(q, K) \succeq (q, KT).$$

The agent never gains when her record is garbled.

(A7) Action data processing. For every $K : A \rightarrow \Delta(O \times R)$, every action processor $S : A \rightarrow \Delta(B)$, every $q \in \Delta(A)$, and every Bayesian pushforward $\widehat{K}$ of K by S at q ,

$$(q, K) \succeq (qS, \widehat{K}).$$

Nor does she gain when the action variable is processed, holding the distribution of visible consequences fixed.

Compounding. For $k \geq 1$, let $P : A \rightarrow \Delta(\{1, \dots, k\})$ be a first-stage record channel and let $(K_1, \dots, K_k)$ be an ordered list of continuation channels $K_i : A \rightarrow \Delta(O \times R)$. Their *compound environment* is

$$(P \triangleright (K_1, \dots, K_k))(o, (i, r) \mid a) := P(i \mid a) K_i(o, r \mid a), \quad i = 1, \dots, k. \quad (2)$$

In words, one action is realised and the environment responds in two stages. First, the action a generates only an interim record i through P ; no payoff is delivered at this stage. Record i selects the continuation channel K_i , through which the same action generates the payoff o and a further record r . The agent does not act again. The interim record remains visible, so the final record is (i, r) . Under q and P , record i is *reached* if $\sum_a q(a) P(i \mid a) > 0$. For every reached record, let q_i denote the posterior distribution of the action.⁷

(A8) Recordwise sure-thing principle. For every $q \in \Delta(A)$, every binary first-stage record channel $P : A \rightarrow \Delta(\{1, 2\})$ for which record 1 is reached, and every three continuation channels $K, L, M : A \rightarrow \Delta(O \times R)$,

$$(q_1, K) \succeq (q_1, L) \iff (q, P \triangleright (K, M)) \succeq (q, P \triangleright (L, M)). \quad (3)$$

Fix q and a binary first-stage record channel P , and suppose record 1 is reached, inducing posterior q_1 . The two compound environments share the continuation M after record 2 and differ only after record 1, where one continues with K and the other with L . The axiom says that the compounds are ranked exactly as the lottery q_1 in channel K and the same lottery in channel L are ranked in their block environment. The common continuation on the other branch cannot affect the comparison. This is a version of Savage's sure-thing principle with the event given by an interim record [42]. Unlike the exogenous event in Savage's formulation, reaching the record can be informative here: $P(1 \mid a)$ may vary with the action. This is why the conditional comparison is evaluated at q_1 . If record 1 is never reached, the axiom is silent, as on Savage's null events. When P is an action-independent public coin, $q_1 = q$, and Axiom (A8) reduces to ordinary mixture independence with the coin retained in the record.

⁶If $(qS)(b) = 0$, the equation leaves that row unrestricted; every statement below applies to every consistent completion, making unreached processed actions irrelevant.

⁷Thus $q_i(a) := q(a) P(i \mid a) / \sum_{a'} q(a') P(i \mid a')$.

3 The representation theorem

For $q \in \Delta(A)$ and $K : A \rightarrow \Delta(O \times R)$, let

$$p_{qK}(o, r) := \sum_a q(a)K(o, r \mid a)$$

be the induced distribution of visible consequences, and define

$$I_{qK}(A; O, R) := \sum_{a, o, r} q(a)K(o, r \mid a) \log \frac{K(o, r \mid a)}{p_{qK}(o, r)}. \quad (4)$$

Summands with $q(a)K(o, r \mid a) = 0$ are set to zero, the logarithm has any fixed base greater than one, and expectations below are taken under the joint distribution $q(a)K(o, r \mid a)$.

Theorem 1. *For a family $\{\succeq_K\}_K$ as described above, the following are equivalent.*

- (i) *The family satisfies Axioms (A1)–(A8).*
- (ii) *There are a nonconstant function $u : O \rightarrow \mathbb{R}$ and a number $\lambda > 0$ such that, for every nonempty finite A, R , every $K : A \rightarrow \Delta(O \times R)$, and every $q, p \in \Delta(A)$,*

$$q \succeq_K p \iff V(q, K) \geq V(p, K), \quad V(q, K) := \mathbb{E}_{qK}[u(O)] + \lambda I_{qK}(A; O, R). \quad (5)$$

Moreover, under these equivalent conditions, block-supported cross-channel comparisons are represented on the same scale: for every finite block environment $\bigsqcup_{k \in I} K_k$, every two distinct blocks $i, j \in I$, and every $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in I} K_k} q_j^j \iff V(q_i, K_i) \geq V(q_j, K_j).$$

We call a family satisfying (A1)–(A8) *trace-tempered*, and likewise the criterion that represents it. Its central restriction is that payoff and trace enter separately. The agent may value good outcomes and may value evidence about her action, but she cannot place an extra value on a good payoff being the revealing consequence. Thus, if under the same lottery two channels give the same expected utility from payoffs and induce the same distribution of posteriors over actions, the agent is indifferent between them, regardless of how payoffs and records are paired (Corollary 34, Appendix C). This separation is derived, not assumed.

3.1 Proof sketch

That the representation satisfies the axioms is a direct verification. The converse has four steps. Appendix B gives the complete proof; Appendix A supplies the pure-trace result used in Step 3.

Step 1: reduction to terminal laws. Fix a full-support q . If two channels induce the same joint distribution of the realised payoff and the posterior over actions, each is a record processing of the other. Block coherence and record processing therefore make them indifferent. Action processing makes action relabellings immaterial and, for an arbitrary q , permits null actions to be deleted. Thus (q, K) is identified by its *terminal law*, the joint distribution of the realised payoff and posterior (Lemma 29(a) and Lemma 28(a)–(b)).

Step 2: cardinalisation. Let a public coin select between two channels. Because the coin is independent of the action, the posterior in either branch remains q . The recordwise sure-thing principle therefore gives mixture independence over terminal laws. With continuity, the mixture-space theorem gives an affine representation for each q . All cardinal content enters here; every other axiom is an ordinal condition, imposed environment by environment (Lemma 29(b)).

Step 3: payoff and trace. On singleton action sets there is no trace, so the affine representation reduces to expected utility. Material relevance makes its index u nonconstant, while action

processing and block coherence make the same payoff scale apply throughout. At the payoff o_* that witnesses trace relevance, the payoff term is constant. Appendix A shows that the remaining order is represented by mutual information. The finite-branch extension of Axiom (A8) (Lemma 10) supplies the chain rule; action processing supplies the corresponding grouping property; and block coherence aligns comparisons across priors. Together these properties give Faddeev’s recursion and hence Shannon entropy [22]. Trace relevance fixes the positive sign, and the common block scale determines a single coefficient λ relative to u (Lemmas 30 and 31).

Step 4: assembly. The finite-branch extension of Axiom (A8) also determines how payoff and trace combine. Continue to use the payoff o_* that witnesses trace relevance. Begin with a compound that pays o_* after every first-stage record. If record y occurs with probability $m(y)$, changing only the payoff after y from o_* to o changes value by $m(y)[u(o) - u(o_*)]$. This baseline compound has value

$$u(o_*) + \lambda I(A; Y).$$

Summing across branches gives the representation when the payoff is determined by the first-stage record. For an arbitrary channel, announce (o, r) as an interim record and then deliver o . This sequential channel and the original channel are record processings of one another and hence are indifferent. The representation follows (Lemmas 32 and 33).

3.2 Uniqueness

A real-valued index W on all finite lottery–channel pairs is a *common-scale representation* if, for every channel $K : A \rightarrow \Delta(O \times R)$ and every $q, p \in \Delta(A)$,

$$q \succeq_K p \iff W(q, K) \geq W(p, K),$$

and, for every finite block environment $\bigsqcup_{k \in I} K_k$, every two distinct blocks $i, j \in I$, and all $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in I} K_k} q_j^j \iff W(q_i, K_i) \geq W(q_j, K_j).$$

Axiom (A8) is ordinal: it says that a common continuation on the other branch cannot affect a conditional comparison. By Lemma 10, this is equivalent—given the other structural axioms—to the requirement that branchwise improvements cannot be reversed by finite compounding. Branch additivity is the corresponding requirement on numerical differences. A common-scale representation W is *branch-additive* if, for every $q \in \Delta(A)$, every $P : A \rightarrow \Delta(\{1, \dots, k\})$, and every two ordered lists of continuation channels $(K_1, \dots, K_k)$ and $(L_1, \dots, L_k)$, with $K_i, L_i : A \rightarrow \Delta(O \times R)$, writing $m(i) := \sum_a q(a)P(i | a)$,

$$W(q, P \triangleright (K_1, \dots, K_k)) - W(q, P \triangleright (L_1, \dots, L_k)) = \sum_{i: m(i) > 0} m(i)[W(q_i, K_i) - W(q_i, L_i)]. \quad (6)$$

A continuation change worth x after record i therefore changes ex-ante value by $m(i)x$, and changes on different branches add.

Proposition 2 (Uniqueness). *Suppose (A1)–(A8) hold, and fix a representation V , with parameters (u, λ) , as in Theorem 1. For any real-valued index W on all finite lottery–channel pairs:*

- (i) *W is a common-scale representation if and only if $W = \varphi \circ V$ for a single strictly increasing $\varphi : [\underline{u}, \infty) \rightarrow \mathbb{R}$, where $\underline{u} := \min_{o \in O} u(o)$.*
- (ii) *A common-scale representation W is branch-additive if and only if $W = \alpha V + \beta$ for some $\alpha > 0$ and $\beta \in \mathbb{R}$.*

The proof is in Appendix C.

Every trace-tempered representation is a common-scale representation and, by the expected-utility and mutual-information chain rules, branch-additive. Part (ii) therefore implies that if (u, λ) and $(\tilde{u}, \tilde{\lambda})$ are two such representations, then

$$\tilde{u} = \alpha u + \beta, \quad \tilde{\lambda} = \alpha \lambda$$

for some $\alpha > 0$ and $\beta \in \mathbb{R}$. Once two non-indifferent payoff outcomes are assigned numerical utilities, λ is unique for the chosen logarithm base. In the pure-trace model of Appendix A, multiplying the index by a positive constant has no behavioural content. Here the payoff normalisation fixes that scale, and λ measures the trade-off between payoff and trace.

3.3 Sign of the trace weight

Uniqueness concerns scale; a complementary observation concerns sign. Axioms (A4), (A6), and (A7) orient the trace component. Write $(A4^-)$, $(A6^-)$, and $(A7^-)$ for the clauses obtained by reversing their displayed comparisons, and write $(A4^0)$, $(A6^0)$, and $(A7^0)$ for the corresponding indifference clauses.

Corollary 3 (Sign trichotomy). *Suppose Axioms (A1)–(A3), (A5), and (A8) hold. Then:*

- (i) *under Axioms (A4), (A6), and (A7), the family has the representation in Theorem 1, with $\lambda > 0$;*
- (ii) *under $(A4^-)$, $(A6^-)$, and $(A7^-)$, it has the same representation with $\lambda < 0$;*
- (iii) *under $(A4^0)$, $(A6^0)$, and $(A7^0)$, it is represented by expected utility, equivalently by the same formula with $\lambda = 0$.*

Thus the trace component has exactly the positive, negative, and neutral orientations.

The proof is in Appendix C.

The directions of the three orientation clauses cannot be chosen independently. For example, under the corollary’s background axioms and any orientation of Axiom (A7), Axiom (A4) is inconsistent with $(A6^-)$. Restrict the two lotteries in Axiom (A4) to their supports and relabel the actions. The second channel is then a total garbling of the first. Axiom $(A6^-)$ weakly prefers the second, whereas Axiom (A4) strictly prefers the first.

The negative case is quiet in unconstrained choice over lotteries. With $\lambda < 0$, every payoff-maximising pure action is optimal. More generally, q is optimal in a fixed channel if and only if it is supported on payoff-maximising actions and $I_{qK}(A; O, R) = 0$. To reveal deniability in optimal choice, one must therefore constrain the lottery or compare channels. Under an imposed fair lottery over two actions, the agent is willing to give up as much as $|\lambda| \log 2$ of expected utility from payoffs to replace full revelation with silence.

3.4 Independence of the axioms

Proposition 4 (Independence of the axioms). *For each $i \in \{1, \dots, 8\}$ there is a family satisfying the other seven axioms and violating Axiom (Ai) .*

The eight families are constructed in Appendix C. Three mark the substantive frontier of the system. A conditional-entropy penalty satisfies every axiom except record data processing and values the removal of extraneous record noise. A Gini posterior-reduction criterion satisfies every axiom except action data processing. The satiation criterion $\mathbb{E}[u(O)] + \min\{I(A; O, R), c\}$, for $c > 0$, satisfies every axiom except the recordwise sure-thing principle and ceases to value trace once c units of information are secured.

4 Choice implications

This section works out what an agent with the representation of Theorem 1 does: which finite designs falsify the model directly (§4.1), when she deliberately randomises (§4.2), what she will pay for changes to the record technology (§4.3), and how that price can be measured and compared across agents (§4.4). Throughout, λ is measured in payoff units per unit of information in the fixed base. Proofs for this section, together with full statements of the supporting results that the text presents in summary form, are collected in Appendix C.3; only one-line verifications are kept here.

4.1 The trace test

The criterion V reads the joint distribution of action and visible consequence, rather than the consequence distribution or action lottery alone. The following two-treatment design separates these objects. In both treatments, the lottery must be knowingly chosen and implemented by the subject, since the primitive ranks her own randomisations.⁸

Proposition 5 (The trace test). *Four actions pay the same outcome o_0 . Two of them leave distinct marks: a_1 always produces record r_1 , and a_2 always produces the distinct record r_2 . The other two are indistinguishable: a_3 and a_4 each produce r_1 or r_2 with equal probability. Call this channel K , and let q' be the fair coin over $\{a_1, a_2\}$ and q'' the fair coin over $\{a_3, a_4\}$.*

- (i) *The two coins induce the same distribution of visible consequences, $p_{q'K} = p_{q''K}$; yet*

$$V(q', K) - V(q'', K) = \lambda \log 2 > 0, \quad \text{so} \quad q' \succ_K q''.$$

- (ii) *Let K° be the channel in which every action pays o_0 and produces r_1 or r_2 with equal probability. Then $q' \sim_{K^\circ} q''$.*

The prediction is strict preference for the revealing coin in the first treatment and exact indifference in the second. Part (i) rules out any criterion that depends only on the distribution of visible consequences. The two coins induce the same distribution over payoff–record pairs, so any such criterion is indifferent between them.

Part (i) restricts only how a theory reads consequences; a theory that also reads the lottery can match it. Existing theories of deliberate randomisation are of this kind: in the hedging model of Cerreia-Vioglio, Dillenberger, Ortoleva, and Riella [15], and in the preference for randomisation documented by Agranov and Ortoleva [2], the value of mixing depends only on the induced distribution and the lottery. Call an index W *channel-blind* if it reads only these two objects: there is a function F , fixed across channels, such that

$$W(q, K) = F(p_{qK}, q).$$

Part (ii) rules these out. Nothing a channel-blind index reads changes between the treatments: in all four lottery–channel combinations the distribution of visible consequences is uniform on $\{(o_0, r_1), (o_0, r_2)\}$, and the lotteries are the same two lotteries. A channel-blind index therefore separates the coins in both treatments or in neither, so none reproduces the pattern of Proposition 5: strict preference under K and indifference under K° .⁹

The first treatment also identifies the orientation of taste: it selects the revealing coin when $\lambda > 0$, the noisy coin when $\lambda < 0$, and indifference when $\lambda = 0$. A strict preference between the coins in the second treatment rejects all three cases of Corollary 3.

⁸A direct value for information about which action was realised is observationally equivalent on this domain. It can be separated by assigning the same lottery externally: the information structure remains, but the subject no longer chooses the lottery.

⁹If W does not depend on how the actions are labelled—an entropy bonus, say—the first treatment is enough: the permutation that swaps a_1 with a_3 and a_2 with a_4 turns q' into q'' and leaves p_{qK} unchanged, so $W(q', K) = W(q'', K)$. The second treatment is needed for label-sensitive indices.

4.2 Deliberate randomisation

Randomisation gives the record something to reveal. A pure action leaves no trace: if only one action can be realised, its consequence cannot reduce uncertainty about it. The agent may therefore give up expected utility to make her action uncertain, so that its consequence can reveal it. Whether she does so depends on how different the actions' visible consequences are. Write $K_a := K(\cdot | a)$ for the distribution of visible consequences under action a and $\bar{u}(a) := \mathbb{E}_{K_a}[u(O)]$ for its expected utility. The Kullback–Leibler divergence $D_{\text{KL}}(P \| Q)$ measures how far P departs from Q .¹⁰

Consider two actions a and b , with $K_a \neq K_b$, and suppose that a has expected-utility advantage $\Delta := \bar{u}(a) - \bar{u}(b) \geq 0$. Proposition 37 in Appendix C.3 gives the exact rule:

$$\delta_a \text{ is optimal} \iff \Delta \geq \lambda D_{\text{KL}}(K_b \| K_a).$$

A small shift of probability from a to b costs Δ per unit. At the pure choice, its marginal trace value is $\lambda D_{\text{KL}}(K_b \| K_a)$. The order in the divergence matters because the question is whether to introduce b into a lottery concentrated on a . If b can produce a consequence that a cannot, the divergence is infinite—the first unit of probability on b is unboundedly informative—and b receives positive probability whatever its finite payoff disadvantage.

Below the threshold the optimal lottery is unique and mixes the two actions. The inferior action receives more probability as λ rises and less as Δ rises. At equal payoffs the chosen mixture maximises the information carried by the channel; in information-theoretic terms, it is the capacity-achieving input.

With more than two actions, each action is compared with the distribution of visible consequences generated by the lottery as a whole.

Proposition 6 (Optimality condition). *Let $\lambda > 0$. A lottery q^* is optimal if and only if there is a constant c such that*

$$\bar{u}(a) + \lambda D_{\text{KL}}(K_a \| p_{q^*K}) = c \quad \text{for } a \in \text{supp}(q^*), \quad \leq c \quad \text{for } a \notin \text{supp}(q^*).$$

An action is credited with its expected utility and with how far its visible consequences depart from those generated on average. At an optimum this total is the same for every action in use; an unused action cannot offer more.

The condition has a sharp support implication. Every visible consequence that can arise under some action occurs with positive probability under any optimal lottery. Hence, if each action can produce some consequence that no other action can produce, every optimal lottery uses every action. No finite payoff disadvantage can eliminate an action with such a private consequence (Proposition 36, Appendix C.3).

The optimal lottery need not be unique, but every optimal lottery induces the same distribution of consequences. If two actions have identical rows, only their combined probability matters: reallocating a fixed total between them changes nothing. Splitting an action into duplicate copies therefore leaves the set of optimal aggregate choices unchanged and permits any division of the original mass among the copies. A model that gives each action its own weight, as multinomial logit does, instead raises the copies' combined share (Proposition 35, Appendix C.3).

Remark 7 (Separated and overlapping traces). If every action leaves its own mark, the row supports are pairwise disjoint, the visible consequence identifies the action, and $I_{qK}(A; O, R) = H(q)$. The unique optimum is then multinomial logit,

$$q^*(a) = \frac{\exp(\bar{u}(a)/\lambda)}{\sum_{a'} \exp(\bar{u}(a')/\lambda)},$$

¹⁰For distributions P and Q , $D_{\text{KL}}(P \| Q) := \sum_{o,r} P(o, r) \log(P(o, r)/Q(o, r))$, with zero-mass summands set to zero. The divergence is $+\infty$ if P assigns positive probability to a consequence and Q assigns zero.

with $1/\lambda$ as the payoff sensitivity when information is measured in natural logarithms. Changing the logarithm base rescales λ . At the other extreme, suppose every action produces the same distribution of payoffs and records. The consequence then reveals nothing about the action, and all actions have the same expected utility. Hence the agent is indifferent among all lotteries. With partial overlap, the ratio of two actions' probabilities can depend on the rest of the menu because adding an action changes the average consequence distribution. Pairwise-disjoint row supports therefore imply independence of irrelevant alternatives, while overlap permits, but does not require, its failure. Remark 38 in Appendix C.3 gives a three-action example in which the ratio falls from 1 to $3/4$. The same logit formula appears in rational inattention for the opposite reason: there the prior is fixed and the channel is chosen; here the channel is fixed and the distribution of actions is chosen [35].

4.3 The price of being on the record

The representation puts different record technologies on a common scale. Processing a record leaves payoffs unchanged, so its price depends only on the information about the action that it removes.

Proposition 8 (Price of a record garbling). *Let $K : A \rightarrow \Delta(O \times R)$, let T be a record processor, and write $L := KT$. Fix $q \in \Delta(A)$ and let (A, O, R, R') carry the joint law $q(a)K(o, r | a)T(o, r' | o, r)$. Then*

$$V(q, K) - V(q, L) = \lambda I(A; R | O, R') \geq 0,$$

with equality if and only if, after (O, R') has been observed, the original record R contains no further information about the action.

Rewriting the record costs λ times the information about the action that the original record carried beyond the rewritten one. A garbling is free exactly when it combines only records that support the same posterior about the action. The agent does not pay for detail as such, only for detail that distinguishes among her actions.¹¹

One application concerns the mixing device itself. Suppose an action-independent coin Y selects K with probability t and L otherwise. If the coin is recorded, the resulting environment is worth $tV(q, K) + (1 - t)V(q, L)$: the agent has no taste for the draw itself. If the coin is hidden, the resulting channel is the rowwise mixture $M_t := tK + (1 - t)L$, and

$$tV(q, K) + (1 - t)V(q, L) - V(q, M_t) = \lambda I(A; Y | O, R) \geq 0.$$

The coin says nothing about the action by itself. Hiding it is costly only when knowing which channel operated would sharpen the inference about the action after the other visible consequences have been observed. The premium is for recording which channel operated, not for drawing the coin. A theory that values mixing as such has no reason to distinguish a recorded draw from an unrecorded one. Corollary 41 in Appendix C.3 gives the full statement.

Uniform rankings do not identify garbling. Suppose two channels K and L generate the same payoff distribution after every action, so they differ only in their records, and every action lottery is worth at least as much under K as under L . This says exactly that K carries at least as much mutual information at every lottery—that K is more capable [32]—but not that L is a garbling of K : two constant-payoff binary channels can be ordered by information at every prior although no record processor maps one into the other (Proposition 43, Appendix C.3). Uniform preference reveals capability, not censorship.

¹¹The same calculation applies to added information. If K^{+S} appends a signal S drawn conditional on (a, o, r) , then $V(q, K^{+S}) - V(q, K) = \lambda I(A; S | O, R)$. Under the deniability case of Corollary 3, the sign is reversed.

4.4 Measuring the trace weight

The coefficient λ is the price of one unit of information in expected-utility units. Any indifference between two environments with different amounts of trace reveals it. Write $E_{qK} := \mathbb{E}_{qK}[u(O)]$.

Proposition 9 (Single information price). *If $(q, K) \sim (p, L)$ and $I_{qK} \neq I_{pL}$, then*

$$\frac{E_{pL} - E_{qK}}{I_{qK} - I_{pL}} = \lambda,$$

and λ is the only number for which this identity holds at every such indifference.

In the information–payoff plane, indifference curves are parallel lines of slope $-\lambda$. The slope does not depend on the prior, the payoff level, the action alphabet, or the amount of information already present. This constancy is the observable restriction that Axiom (A8) adds to the rest of the system. A taste for trace that satiates instead has a measured slope that falls to zero beyond the satiation point.

Consider a lottery q with $H(q) > 0$ and two channels with the same constant payoff, one that reveals the realised action and one that is silent. Revelation is worth $\lambda H(q)$. Thus compensating the silent channel until the agent is indifferent measures her willingness to pay for trace. One implementation dilutes both channels by the same public side branch, reached with probability ε . Normalise the two outcomes in Axiom (A3) so that $u(o^+) = 1$ and $u(o^-) = 0$. On the revealing side the branch pays o^- for sure; on the silent side it pays o^+ with probability β and o^- otherwise. If β produces indifference, then

$$\lambda = \frac{\varepsilon}{1 - \varepsilon} \frac{\beta}{H(q)}.$$

The common baseline payoff’s utility cancels, so it need not be known. Proposition 45 and Remark 50 in Appendix C.3 give the construction, boundary cases, and controls.

Varying the prior tests the model rather than merely estimating λ : the premia at $q = (1/2, 1/2)$ and $q = (1/4, 3/4)$ must stand in the ratio

$$\frac{\log 2}{\frac{1}{4} \log 4 + \frac{3}{4} \log \frac{4}{3}} \approx 1.23.$$

A quadratic posterior-reduction criterion gives $4/3$. Agreement across the two designs overidentifies λ ; disagreement rejects the Shannon specification even if each design separately admits an information price. The switching threshold in §4.2 supplies a third estimate of λ , from optimal choices rather than stated indifferences: the payoff advantage at which the agent abandons mixing equals λ times a known divergence. It must return the same number.

Within a fixed channel, λ governs the demand for trace. Holding payoff utility fixed, raising λ leads the agent to choose a lottery that carries weakly more information about her action and yields weakly less expected utility from payoffs. As λ approaches zero, payoff is maximised first and trace breaks ties; as λ becomes large, trace is maximised first and payoff breaks ties (Proposition 39 and Corollary 40, Appendix C.3).

The coefficient also permits comparisons across agents. Say agent 2 is more trace-seeking than agent 1 if she weakly prefers an environment to a silent benchmark whenever agent 1 does. This ordering holds exactly when their representations can use the same payoff utility, after a common normalisation, with $\lambda_2 \geq \lambda_1$. Agents with different payoff tastes are not ordered (Proposition 48, Appendix C.3; compare [24]).

5 Related literature

This paper intersects several literatures that differ in what is ranked, which way information flows, and whether an information functional is assumed or derived. In most models of information, a

signal tells the agent about an exogenous state; here a visible consequence carries information about the agent’s realised action.

Kreps [33] and Dekel, Lipman, and Rustichini [18] use preferences over menus to represent flexibility and uncertainty about future tastes; Pattanaik and Xu [38] rank opportunity sets by the freedom they provide. Gul and Pesendorfer [28] use the same domain to separate temptation, commitment, and self-control, while Kreps and Porteus [34] use temporal lotteries to study preferences over the timing of uncertainty resolution. In this paper, a choice problem fixes the feasible actions, the decision rights, and the channel from actions to visible consequences. The primitive preference ranks lotteries over actions and therefore reads the dependence between the realised action and its visible consequence, rather than the size of a menu, the value of commitment, or the timing of resolution.

Grant, Kajii, and Polak [27] characterise an intrinsic preference for information independent of its use in planning; Caplin and Leahy [14] model anticipatory feelings generated by beliefs, while Ely, Frankel, and Kamenica [21] study preferences over paths of beliefs and the resulting value of suspense and surprise. These papers show that information need not be instrumental, but the relevant beliefs concern uncertainty faced by the agent, whereas the posterior here concerns her realised action. The axioms also restrict its value to expected entropy reduction and fix its rate of substitution with material utility.

Cerreia-Vioglio, Dillenberger, Ortoleva, and Riella [15] study deliberate randomisation and characterise a model in which mixing hedges uncertainty about how to evaluate lotteries; Agranov and Ortoleva [2] document stochastic choice under announced repetition and evidence that much of it is deliberate. Trace-tempered choice can also make a nondegenerate lottery optimal. Proposition 5, however, holds fixed the action lotteries and their distributions of visible consequences while changing whether the consequence identifies the realised action; its second treatment removes that dependence. Together the two treatments rule out every channel-blind criterion that reads only the lottery and its consequence distribution. Randomisation is an implication of traceable agency, not its primitive motive.

In signaling models following Spence [45], actions convey information about hidden characteristics. Glazer and Konrad [26] study charitable giving as a signal of wealth, and Andreoni and Bernheim [3] study social image and audience effects. Bénabou and Tirole [9] allow concerns for social reputation or self-respect. Their earlier paper [8] studies self-serving beliefs about ability, while Bernheim [10] shows how status concerns can induce agents with heterogeneous preferences to choose the same action, making behaviour less informative about predispositions—an instrumental counterpart to the deniability case in Corollary 3. In these models, inference matters through esteem, sanction, reputation, or self-regard. Here no such response enters utility. Beliefs serve only as the statistical yardstick for how clearly the consequence identifies the act.

The proof also relates to axiomatic characterisations of entropy. Shannon [43] characterised entropy from continuity, monotonicity on uniform distributions, and a grouping recursion; Faddeev [22] reduced these to symmetry, continuity of the binary entropy, and the recursion itself. The proof uses the strong-additivity form of Faddeev’s theorem given by Baez, Fritz, and Leinster [7]; other formulations and extensions include Khinchin [30], Rényi [40], Aczél and Daróczy [1], and Csiszár [17]. Mutual information can also be obtained through relative entropy, which Baez and Fritz [5] characterise through properties of Bayesian inference.

In those characterisations, a numerical information functional is primitive and structural conditions are imposed directly on it; here the primitive is an ordinal family of weak orders over action lotteries. The information chain rule and Faddeev’s recursion are consequences of behavioural axioms, not postulates on a given function. The theorem also calibrates the information term against material utility. The result is not another characterisation of entropy; it derives entropy reduction as the representation of a preference over trace.

Blackwell [11] compares experiments by their value in all downstream decision problems; Gilboa and Lehrer [25] characterise which functions on partitions can arise as values of information

in Bayesian decision problems, and Azrieli and Lehrer [4] extend the analysis to stochastic information structures. Cabrales, Gossner, and Serrano [12] obtain entropy reduction from the decisions of ruin-averse investors. For Blackwell’s decision-maker, information is an input: she can use a finer signal or ignore it, so more information cannot hurt. Here information is an output: the record is evidence about the agent, and no decision follows it. Uniform preference between two record technologies identifies the more-capable order, not Blackwell garbling (Proposition 43, Appendix C.3). Whether a finer record is desirable is therefore a matter of taste—positive for the trace-seeker, negative under the deniability dual of Corollary 3, and zero under expected utility. Being better informed is a theorem; being better known is a preference.

The closest formal comparison is with rational inattention and information costs. Sims [44] imposes a finite Shannon-capacity constraint on information flow, while Matějka and McKay [35] show that a cost proportional to mutual information yields a generalized multinomial-logit rule. Rational inattention fixes a prior over states and allows the agent to choose how much to learn before acting, whereas here the channel from actions to consequences is fixed and the agent chooses a distribution over actions. Pairwise-disjoint row supports yield multinomial logit. With overlapping rows, logit can still obtain, but independence of irrelevant alternatives can fail, as Remark 7 shows.

Caplin, Dean, and Leahy [13] characterise Shannon rational inattention through Invariance under Compression and develop posterior-separable generalisations; Denti [19] gives testable conditions under which state-dependent stochastic choice is rationalised by posterior-separable information costs. Frankel and Kamenica [23] characterise measures of information that arise as the value of news in a decision problem and show that the corresponding expected reduction in uncertainty equals expected information. In these papers, the information cost or the underlying optimisation problem supplies the cardinal scale; the present primitive is ordinal, and the common scale for payoff and trace is derived.

Pomatto, Strack, and Tamuz [39] take a cardinal, prior-independent cost functional on Blackwell experiments and show that monotonicity, continuity, product additivity, and constant marginal cost characterise weighted sums of Kullback–Leibler divergences between state-conditional signal distributions. Their costs and the trace value studied here are not nested. The trace component $(q, K) \mapsto I_{qK}(A; O, R)$ depends on the prior and is not a prior-free additive cost of an experiment. Full revelation has finite trace value $H(q)$; under every nonzero cost in their class it is excluded from the finite-cost domain and would have infinite cost. Record and action data processing are behavioural assumptions about whether the agent values evidence of her action; reversing them, with the taste for revelation, yields a preference for deniability, not an information-acquisition cost. The information-cost literature prices learning about the world; the present paper prices being learned about.

6 Interpretation and scope

Control and empowerment. The information term reads in two directions. Backward, it measures trace: how much the visible consequence reveals about the realised action. Forward, it measures control: how strongly the action shapes the distribution of visible consequences. Mutual information is symmetric, but the channel is directional— K maps actions into distributions over consequences—so the two readings describe the same dependence. No choice among lotteries separates them. The control in question is informational: how distinguishable actions are in their consequences, not the agent’s power to obtain the consequences she prefers.

The forward reading has an exact counterpart. Suppose $\mathbb{E}[u(O) \mid a] = \bar{u}$ for every action. Since $\lambda > 0$, maximising V over q gives

$$\max_{q \in \Delta(A)} V(q, K) = \bar{u} + \lambda C(K), \quad C(K) := \max_{q \in \Delta(A)} I_{qK}(A; O, R),$$

the Shannon capacity of the channel. When the visible consequence is read as the agent’s sensor state, $C(K)$ is the one-step empowerment index of Klyubin, Polani, and Nehaniv [31]; see also [41]. Under this supposition, the trace-tempered agent solves exactly the problem that defines one-step empowerment: her optimal lotteries are the capacity-achieving action distributions, and the size of λ plays no role in the ranking—the price is visible only against payoffs. Empowerment summarises the channel by the optimised value $C(K)$; the representation recovers the ranking beneath it, $q \mapsto I_{qK}(A; O, R)$, from preferences and prices it against material utility. Empowerment is thus the trace part of indirect utility when expected payoffs are equal; once they differ across actions, the ranking also reflects their material value, which the empowerment index does not encode.

Evidence, not authorship. The trace term prices evidence of agency, not authorship as such. A pure action has zero trace value even when its consequence names it perfectly: an act settled in advance leaves no uncertainty for the consequence to resolve. A coin the agent flips herself can do better. The model does not say that surrendering a choice to chance creates more agency; it says that an unsettled act can leave more evidence than a predetermined one. The lottery must be hers—knowingly chosen and implemented, as in the test of §4.1. This is the Underground Man’s problem: the proof he wants cannot come from an act fixed in advance.

What the axioms preclude. A principal substantive exclusion is selective legibility. Mutual information responds to an act’s probability and to the distinctiveness of its consequences, not to what the act is. An agent who wants her generous acts recognised and her mean acts hidden may therefore prefer an informative record in one channel and a garbling in another. Such preferences violate the global data-processing requirement in Axiom (A6) and fall outside all three cases of the sign trichotomy. They are not irrational; they are simply not represented here. A model of selective legibility would have to make the value of evidence depend on its content, not only on the amount of uncertainty it removes.

The payoff–record distinction. Although the information term treats O and R jointly, the axioms cannot. Axiom (A6) permits the record to be garbled while the payoff is held fixed; this separates a change in evidence from a change in material consequences. Axiom (A8) permits records to arrive in stages while the payoff is delivered once. Its recordwise sure-thing restriction makes branchwise comparisons commensurable and supplies the cardinal structure used to place payoff utility and information on a common scale. Thus either O or R may reveal the action, but only O enters material utility. The record is also assumed to have no utility beyond what it reveals: a record that is itself pleasant or shameful belongs in O , or the domain must be enlarged. Once u is normalised, willingness to sacrifice payoff utility for additional mutual information identifies λ .

One shot, with no audience. The model describes one episode. It specifies neither an audience nor any response to the beliefs induced by the record; the posterior is a statistic of the implemented lottery and the channel, not anyone’s belief. The stages in Axiom (A8) divide a single interaction: they create neither a reputational relationship nor a feedback policy, so dynamic empowerment and the value of keeping future options open lie outside the model. An experiment could vary whether an audience observes the record while holding the agent’s payoffs and information fixed. A response to that treatment would reveal an external component at work—praise, sanction, reputation; where such responses change material consequences, they belong in O and the channel. Persistence in private would be consistent with an internal mechanism, such as self-image, but would not identify it. Distinguishing these sources requires additional primitives.

Proofs

A Pure trace on the constant-payoff domain

A.1 Binary and finite branching

We first record the finite-branch form of Axiom (A8) used throughout the proofs. The result also makes explicit where the other structural axioms enter the equivalence.

Lemma 10 (Finite-branch extension). *Suppose Axioms (A1) and (A5)–(A7) hold. Then Axiom (A8) is equivalent to the following property: for every $k \geq 1$, every first-stage record channel $P : A \rightarrow \Delta(\{1, \dots, k\})$, every $q \in \Delta(A)$, and every two ordered lists of continuation channels $(K_1, \dots, K_k)$ and $(L_1, \dots, L_k)$, with $K_i, L_i : A \rightarrow \Delta(O \times R)$, if*

$$(q_i, K_i) \succeq (q_i, L_i) \quad \text{for every reached } i,$$

then

$$(q, P \triangleright (K_1, \dots, K_k)) \succeq (q, P \triangleright (L_1, \dots, L_k)), \quad (7)$$

strictly if one reached branch comparison is strict.

Proof. Assume first Axiom (A8). Put

$$m(i) := \sum_a q(a) P(i \mid a), \quad I^+ := \{i \in \{1, \dots, k\} : m(i) > 0\}.$$

If $m(i) = 0$, then $P(i \mid a) = 0$ for every $a \in \text{supp } q$. Consequently, changing the continuation after such a record changes the compound channel only on action rows outside $\text{supp } q$. Apply Axiom (A7) in both directions using the identity action processor: its Bayesian pushforwards agree on supported rows and allow arbitrary completions elsewhere. Continuations following unreached records are therefore immaterial. We may hold them fixed and enumerate the reached records as $I^+ = \{i_1, \dots, i_n\}$.

If $n = 1$, the first-stage record is constant on $\text{supp } q$. Deleting or restoring that constant record is an invertible record relabelling, up to rows outside $\text{supp } q$; Axioms (A6) and (A7) therefore identify the compound comparison with the sole branch comparison. Suppose henceforth that $n \geq 2$.

For $j = 0, \dots, n$, define hybrid continuation channels by

$$H_i^j := \begin{cases} K_i, & i \in \{i_1, \dots, i_j\}, \\ L_i, & i \in I^+ \setminus \{i_1, \dots, i_j\}, \end{cases}$$

and use a common arbitrary continuation on $\{1, \dots, k\} \setminus I^+$. Write $E_j := P \triangleright (H_1^j, \dots, H_k^j)$. Fix $j \geq 1$ and collapse the first-stage record to the binary channel

$$B_j(1 \mid a) := P(i_j \mid a), \quad B_j(2 \mid a) := \sum_{i \neq i_j} P(i \mid a).$$

Both binary records are reached. Put all binary continuations on the common record set $\{1, \dots, k\} \times R$. Pad the two continuations that may differ by

$$\widetilde{K}_j(o, (i, r) \mid a) := \mathbf{1}_{\{i=i_j\}} K_{i_j}(o, r \mid a), \quad \widetilde{L}_j(o, (i, r) \mid a) := \mathbf{1}_{\{i=i_j\}} L_{i_j}(o, r \mid a).$$

On record 2, collect all unchanged branches in the continuation channel

$$M_j(o, (i, r) \mid a) := \frac{P(i \mid a)}{B_j(2 \mid a)} H_i^j(o, r \mid a), \quad i \neq i_j, \quad (8)$$

with zero probability on records (i_j, r) , whenever $B_j(2 \mid a) > 0$, and complete the other rows arbitrarily. Since $H_i^j = H_i^{j-1}$ for $i \neq i_j$, this continuation is common to the two comparisons. Binary record 1 has probability $m(i_j)$ and carries the conditional action lottery q_{i_j} ; record 2 carries the complementary conditional lottery, and their mixture is q . Appending or deleting the constant label i_j is an invertible record relabeling, so Axiom (A6) identifies the comparison of $\widetilde{K}_j$ and $\widetilde{L}_j$ with that of K_{i_j} and L_{i_j} . Hence Axiom (A8) gives

$$(q, B_j \triangleright (\widetilde{K}_j, M_j)) \succeq (q, B_j \triangleright (\widetilde{L}_j, M_j)),$$

strictly whenever the comparison at q_{i_j} is strict.

The nested records $(1, (i_j, r))$ and $(2, (i, r))$ correspond bijectively to the flat records (i, r) through

$$(1, (i_j, r)) \mapsto (i_j, r), \quad (2, (i, r)) \mapsto (i, r) \quad (i \neq i_j).$$

This relabelling is a payoff-preserving record processing in both directions. Axiom (A6) therefore makes the two binary compounds indifferent to E_j and E_{j-1} , respectively. Thus $(q, E_j) \succeq (q, E_{j-1})$, strictly at any strict replacement. Place the finite collection $E_0, \dots, E_n$ in one block environment. Axiom (A5) transports each pairwise comparison to that environment, and transitivity in Axiom (A1) yields $(q, E_n) \succeq (q, E_0)$, strictly if at least one step is strict. Restoring the immaterial null branches proves (7).

Conversely, assume the displayed finite-branch property, including its strict clause. In the binary environment of Axiom (A8), reflexivity and duplication coherence make the common continuation weakly comparable to itself, so the finite-branch property gives the forward implication in (3). If the aggregate comparison in that equation held while $(q_1, K) \not\succeq (q_1, L)$, completeness would give $(q_1, L) \succ (q_1, K)$. Applying the strict finite-branch property to the reversed lists would give the strict reverse aggregate comparison, a contradiction. This proves the converse implication and hence Axiom (A8). $\square$

Corollary 11 (One-branch insertion). *Under the hypotheses of Lemma 10, fix $q \in \Delta(A)$, a first-stage record channel $P : A \rightarrow \Delta(\{1, \dots, k\})$, a reached record i^* , and continuation channels $K_1, \dots, K_k, L_1, \dots, L_k : A \rightarrow \Delta(O \times R)$ such that $K_i = L_i$ for every $i \neq i^*$. Then*

$$(q_{i^*}, K_{i^*}) \succeq (q_{i^*}, L_{i^*}) \iff (q, P \triangleright (K_1, \dots, K_k)) \succeq (q, P \triangleright (L_1, \dots, L_k)),$$

and the same equivalence holds for strict preference.

Proof. The forward implication follows from Lemma 10, using duplication coherence on the common branches. If the aggregate comparison held but the branch comparison failed, completeness would make the reversed branch comparison strict; the strict part of the finite-branch extension would then give a strict reverse aggregate comparison. The strict equivalence follows by applying the weak equivalence in both directions. $\square$

In the appendices, references to Axiom (A8) with more than two first-stage records mean its finite-branch extension in Lemma 10; when only one branch changes, we cite Corollary 11.

The relevance axioms in the main text are stated inside fixed channels. The proofs below use their equivalent comparison-environment forms.

For $o \in O$, write $K_o : \{*\} \rightarrow \Delta(O \times \{*\})$ for the channel that delivers o for sure. On a finite action set A , write $K_o^{\text{id}} : A \rightarrow \Delta(O \times A)$ for the channel that delivers o and reports the action, and $K_o^0 : A \rightarrow \Delta(O \times \{*\})$ for the channel that delivers o and no record.

Lemma 12 (Fixed-channel and environment relevance). *Given Axioms (A1), (A5), (A6), (A7), and (A8), the following equivalences hold.*

- (a) *Axiom (A3) is equivalent to the existence of $o^+, o^- \in O$ such that*

$$(\delta_*, K_{o^+}) \succ (\delta_*, K_{o^-}).$$

- (b) *Axiom (A4) is equivalent to the following condition: there is an outcome $o_* \in O$ such that, for every finite A with $|A| \geq 2$ and every full-support $q \in \Delta(A)$,*

$$(q, K_{o_*}^{\text{id}}) \succ (q, K_{o_*}^0).$$

Proof. For part (a), apply action processing to either pure lottery in the two-action channel of Axiom (A3). Collapsing its support to a singleton gives K_{o_+} or K_{o_-} . Conversely, the singleton can be processed back to the corresponding action; the unrestricted row at the unreached action completes the pushforward to the original channel. Thus each pure lottery is indifferent to the corresponding singleton experiment. Weak order and block coherence transport the strict comparison in either direction.

For part (b), let o_* witness Axiom (A4) and first restrict its two lotteries to their supports. Action processing in both directions identifies the first with fair binary full revelation and the second with a fair binary channel whose two rows are the same. Record processing in both directions identifies the latter channel with silence. Hence the fixed-channel axiom is equivalent to

$$((\frac{1}{2}, \frac{1}{2}), K_{o_*}^{\text{id}}) \succ ((\frac{1}{2}, \frac{1}{2}), K_{o_*}^0) \quad (9)$$

on a binary action set.

Now let p have full support on a binary set. Choose $0 < \varepsilon \leq 2 \min_i p_i$ and an interim record y with $P(y | i) = \varepsilon / (2p_i)$. The branch has probability ε and posterior $(\frac{1}{2}, \frac{1}{2})$. Compare two continuation plans: after that branch the first reveals the action and the second is silent; after every other branch both are silent. Equation (9) and Corollary 11 give a strict comparison of the two compounds. Full revelation can be garbled into the first compound, and the second compound can be garbled into silence. Record processing and transitivity therefore give $(p, K_{o_*}^{\text{id}}) \succ (p, K_{o_*}^0)$.

Finally, let q have full support on any finite A with $|A| \geq 2$, and partition A into two nonempty cells. The channel that reports only the cell is, by action processing in both directions, indifferent to binary full revelation under the induced binary prior; silence on A is similarly indifferent to binary silence. The cell-reporting channel is therefore strictly preferred to silence, and full revelation is weakly preferred to the cell report by record processing. This proves the displayed comparison for every q and A at o_* . The converse follows by taking a fair binary prior and reversing the support identifications used above. $\square$

Whenever Axiom (A4) is imposed below, fix one of its witnessing outcomes and denote it by o_* . Every channel between nonempty finite sets $P : A \rightarrow \Delta(X)$ has the constant-payoff lift

$$K_P(o, x | a) := \mathbf{1}_{\{o=o_*\}} P(x | a). \quad (10)$$

Throughout this appendix, comparisons of pure record channels mean comparisons of these lifts in the model already defined in the main text:

$$q \succeq_P q' \iff q \succeq_{K_P} q'.$$

We similarly write $(q, P) \succeq (r, Q)$ for the main-text two-block comparison of (q, K_P) and (r, K_Q) . Likewise, $P \sqcup Q$, block-supported lotteries, action and record processing, ordered lists of continuation channels, and compounds $P_1 \triangleright (Q_1, \dots, Q_k)$ are shorthand for the corresponding main-text constructions after applying (10). The lift commutes with each construction up to canonical tagged or singleton-record bijections; those bijections are neutral by Axiom (A6) applied in both directions. No new primitive relation or behavioural condition is introduced here.

Two pieces of notation will be used repeatedly. If $S : A \rightarrow \Delta(B)$ is an action processor, let $S^q P : B \rightarrow \Delta(X)$ denote any Bayesian pushforward satisfying

$$(qS)(b)(S^q P)(x | b) = \sum_a q(a) S(b | a) P(x | a).$$

Its rows at zero-mass reports are arbitrary, exactly as in (1). For channels $P : A \rightarrow \Delta(X)$ and $Q : B \rightarrow \Delta(Y)$, their independent product is

$$(P \otimes Q)((x, y) \mid (a, b)) := P(x \mid a)Q(y \mid b).$$

For $q \in \Delta(A)$ and $P : A \rightarrow \Delta(X)$, write

$$m_{qP}(x) := \sum_a q(a)P(x \mid a), \quad r_x^{qP}(a) := \frac{q(a)P(x \mid a)}{m_{qP}(x)} \quad \text{when } m_{qP}(x) > 0,$$

and define the finite posterior law

$$\mu_{qP} := \sum_{x: m_{qP}(x) > 0} m_{qP}(x) \delta_{r_x^{qP}}.$$

Let Id_A be full revelation and U_A the one-record uninformative channel. With logarithms in the fixed base used in the main text, put

$$I_{qP}(A; X) := H(m_{qP}) - \sum_a q(a)H(P(\cdot \mid a)) = H(q) - \int_{\Delta(A)} H(r) d\mu_{qP}(r).$$

Proposition 13 (Pure-trace lemma). *Suppose the preference family satisfies main-text Axioms (A1), (A2), and (A4)–(A8). Then, for every record channel between nonempty finite sets $P : A \rightarrow \Delta(X)$ and every $q, q' \in \Delta(A)$,*

$$q \succeq_P q' \iff I_{qP}(A; X) \geq I_{q'P}(A; X).$$

The same numerical scale represents every block-supported comparison: for every nonempty finite family $\{P_i : A_i \rightarrow \Delta(X_i)\}_{i \in J}$ of channels between nonempty finite sets, distinct $i, j \in J$, and priors $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in J} P_k} q_j^j \iff I_{q_i P_i}(A_i; X_i) \geq I_{q_j P_j}(A_j; X_j).$$

A.2 Proof

Assume the hypotheses of Proposition 13. Every appeal below to weak order, continuity, block coherence, record processing, action processing, the finite-branch extension of the recordwise sure-thing principle, or positive trace orientation is an appeal to Axiom (A1), (A2), (A5), (A6), (A7), (A8), or (A4), respectively, on the constant-payoff lifts. Thus the proof has no auxiliary axiom system.

How to read the proof. The argument has five interfaces. Each stage ends with exactly the object needed by the next one:

- Stage 1: replace channels by Bayes-plausible posterior laws, derive public-mixture independence, and cardinalise each fixed-prior quotient using only the segment closedness supplied by primitive continuity;
- Stage 2: make relabelling, undetectable distinctions, and restriction to the positive support exact consequences of action processing;
- Stage 3: transport those affine values through independent products and into one cross-prior block scale, allowing provisionally a bilinear interaction;
- Stage 4: identify reached-branch coefficients, prove their cocycle, and align the scales on nested support faces;

Stage 5: compare product and sequential decompositions, eliminate the interaction, and invoke Faddeev's recursion to obtain Shannon entropy and hence mutual information.

The product-coefficient and branch tangent-space calculations are the two technical subarguments. A reader interested in the logical spine may use their lemma statements as interfaces. Product calibration is placed before branch calibration because this makes the scalar algebra easier to read. The early full-revelation normalisation and exact relabelling lemma below supply the coherence needed by that order of presentation.

A.2.1 Stage 1: posterior-law quotient and affine fibre values

For a finite action set A , let X_A be the set of finitely supported probability measures on $\Delta(A)$, and let

$$b(\mu) := \int_{\Delta(A)} s \, d\mu(s), \quad \mathcal{M}_q := \{\mu \in \mathsf{X}_A : b(\mu) = q\}.$$

Thus $\mathcal{M}_q$ is the mixture set of finite Bayes-plausible posterior laws with prior q , and $\mu_{qP} \in \mathcal{M}_q$ for every finite channel P .

Lemma 14 (Coherent comparisons across blocks). *The relation on prior-channel pairs defined above is complete and transitive. It is unaffected by adding or deleting blocks other than the two being compared. In particular, for fixed q the induced relation on channels is a weak order, and*

$$q \succeq_P r \iff (q, P) \succeq (r, P).$$

For fixed alphabets, its graph is closed under entrywise convergence of the two channels and convergence of the two priors.

Proof. Completeness is Axiom (A1) in the two-block environment. To prove transitivity, suppose $(q_1, P_1) \succeq (q_2, P_2)$ and $(q_2, P_2) \succeq (q_3, P_3)$. Put the three pairs in the common environment $P_1 \sqcup P_2 \sqcup P_3$. Axiom (A5) transports the two hypotheses to that environment, Axiom (A1) composes them there, and Axiom (A5) deletes the middle block. Applying Axiom (A5) with the ordered pair of block labels reversed gives

$$(q_2, P_2) \succeq (q_1, P_1) \iff q_2^1 \succeq_{P_1 \sqcup P_2} q_1^0,$$

so reversal is also canonical. The displayed within-channel equivalence is the duplication clause of Axiom (A5). Finally, a block channel is a continuous affine function of its component channels. Closedness on fixed alphabets is therefore exactly Axiom (A2) applied to the common block environment. $\square$

Lemma 15 (Posterior-law quotient). *Fix a full-support $q \in \Delta(A)$. If finite channels $P : A \rightarrow \Delta(X)$ and $Q : A \rightarrow \Delta(Y)$ satisfy*

$$\mu_{qP} = \mu_{qQ},$$

then they are mutually obtainable by record post-processing. Consequently

$$(q, P) \sim (q, Q),$$

and either channel may be substituted for the other in every block comparison at prior q . Hence the fixed- q order of channels descends to a well-defined weak order $\succeq_q$ on $\mathcal{M}_q$.

Proof. We give the finite matching argument because it also handles repeated and zero-probability records. Put

$$m_P(x) := \sum_a q(a)P(x \mid a), \quad m_Q(y) := \sum_a q(a)Q(y \mid a).$$

For every posterior s occurring with positive probability, let

$$X_s := \{x : m_P(x) > 0, r_x^{qP} = s\}, \quad Y_s := \{y : m_Q(y) > 0, r_y^{qQ} = s\}.$$

Equality of posterior laws gives the common positive mass

$$\lambda_s := \sum_{x \in X_s} m_P(x) = \sum_{y \in Y_s} m_Q(y).$$

For $x \in X_s$, define a stochastic record kernel by

$$T(y | x) := \begin{cases} m_Q(y)/\lambda_s, & y \in Y_s, \\ 0, & y \notin Y_s. \end{cases}$$

Choose arbitrary stochastic rows at records x with $m_P(x) = 0$. Full support of q implies that such a record has $P(x | a) = 0$ for every a .

If $y \in Y_s$, Bayes' rule gives

$$\sum_{x \in X_s} P(x | a) = \frac{\lambda_s s(a)}{q(a)}, \quad Q(y | a) = \frac{m_Q(y) s(a)}{q(a)}.$$

It follows that

$$(PT)(y | a) = \sum_{x \in X_s} P(x | a) \frac{m_Q(y)}{\lambda_s} = Q(y | a).$$

Both sides vanish when $m_Q(y) = 0$, again by full support. Thus $PT = Q$. Interchanging P and Q constructs T' with $QT' = P$.

Axiom (A6) applied to T and T' yields the two weak comparisons; Lemma 14 puts the reverse comparison into the same block order and gives indifference. If P, P' have the same posterior law, place $(q, P), (q, P')$ and the two objects being compared in one finite block environment. Transitivity then permits replacement of P by P' on either side. The quotient order is consequently independent of the implementing channel and inherits completeness and transitivity from Lemma 14. $\square$

Every element of $\mathcal{M}_q$ is implementable when q has full support. Indeed, if

$$\mu = \sum_{k=1}^n \lambda_k \delta_{s_k}, \quad \lambda_k > 0, \quad \sum_k \lambda_k s_k = q,$$

then

$$C_\mu(k | a) := \frac{\lambda_k s_k(a)}{q(a)} \tag{11}$$

defines a channel and $\mu_{qC_\mu} = \mu$.

For channels $P : A \rightarrow \Delta(X)$ and $R : A \rightarrow \Delta(Z)$ and $t \in [0, 1]$, let $tP \oplus (1-t)R$ be the publicly tagged mixture: it reports which component was selected as well as that component's record. Its posterior law is

$$\mu_{q, tP \oplus (1-t)R} = t\mu_{qP} + (1-t)\mu_{qR}. \tag{12}$$

Lemma 16 (Public-mixture independence). *Let q have full support. For all finite channels P, Q, R on the action set A and every $t \in (0, 1)$,*

$$(q, P) \succeq (q, Q) \iff (q, tP \oplus (1-t)R) \succeq (q, tQ \oplus (1-t)R).$$

Equivalently, the quotient order on $\mathcal{M}_q$ satisfies mixture independence.

Proof. If necessary, embed the record alphabets of P, Q, R into one tagged disjoint union and fill the unused coordinates with zeros. This does not change any posterior law, so Lemma 15 makes the padding neutral.

Let $B : A \rightarrow \Delta(\{1, 2\})$ be the uninformative first-stage channel with $B(1 \mid a) = t$ for every a . Both branches are reached, and their posterior is q . The compounds

$$B \triangleright (P, R), \quad B \triangleright (Q, R)$$

have the posterior laws on the right-hand side of (12); they differ from the displayed public mixtures only by record labels. Axiom (A8) gives the desired biconditional directly. Posterior-law substitution transfers it between the compounds and the tagged mixtures. $\square$

Lemma 17 (Affine posterior-law representation). *For every full-support $q \in \Delta(A)$ there is an affine functional*

$$F_q : \mathcal{M}_q \longrightarrow \mathbb{R}$$

such that, for all finite channels P, Q on A ,

$$(q, P) \succeq (q, Q) \iff F_q(\mu_{qP}) \geq F_q(\mu_{qQ}). \quad (13)$$

If $|A| \geq 2$, F_q is unique up to a positive affine transformation. If $|A| = 1$, $\mathcal{M}_q$ is a singleton and we take $F_q \equiv 0$.

Proof. By Lemma 15 and (11), $\succeq_q$ is a weak order on the mixture set $\mathcal{M}_q$. Lemma 16 gives mixture independence. It remains only to verify Herstein–Milnor’s segment-continuity condition. This is where primitive Axiom (A2) is used; no topology on the collection of all finite posterior laws is needed.

Fix $\mu, \nu, \rho \in \mathcal{M}_q$ and choose finite implementing channels P, Q, R . For $t \in [0, 1]$ define

$$M_t := tP \oplus (1 - t)Q.$$

All M_t have the single fixed record alphabet given by the tagged union of the record alphabets of P and Q . If $t_n \rightarrow t$, then $M_{t_n} \rightarrow M_t$ entrywise. After adjoining the fixed block R , all comparisons occur in one fixed finite action and record alphabet. Axiom (A2) therefore makes both sets

$$\{t \in [0, 1] : M_t \succeq_q R\}, \quad \{t \in [0, 1] : R \succeq_q M_t\}$$

closed. By (12), these are exactly

$$\{t : t\mu + (1 - t)\nu \succeq_q \rho\}, \quad \{t : \rho \succeq_q t\mu + (1 - t)\nu\}.$$

Thus the Herstein–Milnor mixture-space theorem [29] applies and supplies an affine representation F_q . When $|A| \geq 2$, Axiom (A4) makes the order nonconstant, so the usual positive-affine uniqueness conclusion applies. For a singleton action set, Bayes plausibility forces every posterior law to equal δ_q . $\square$

The preceding proof uses only closedness along each particular public-mixture segment. It neither assumes nor proves closedness under arbitrary weakly convergent sequences whose support sizes grow. It also requires no pointwise integrand on $\Delta(A)$: every subsequent calculation uses only the affinity of F_q on $\mathcal{M}_q$ and its linear part on the affine hull.

Let

$$\delta_q \in \mathcal{M}_q$$

be the posterior law of the uninformative channel and let

$$\chi_q := \sum_{a \in A} q(a)\delta_{e_a} \in \mathcal{M}_q$$

be the posterior law of full revelation.

Lemma 18 (Canonical fibre normalization). *The representatives in Lemma 17 may be chosen so that*

$$F_q(\delta_q) = 0, \quad F_q(\mu) \geq 0 \quad (\mu \in \mathcal{M}_q), \quad (14)$$

and, whenever $|A| \geq 2$,

$$F_q(\chi_q) = 1. \quad (15)$$

These two anchors select a unique affine representative on every nondegenerate full-support fibre. On a singleton fibre we retain $F_q \equiv 0$.

Proof. Every channel P can be record-postprocessed to the uninformative channel by deleting its record. Axiom (A6) and (13) therefore imply

$$F_q(\mu_{qP}) \geq F_q(\delta_q).$$

Every law in $\mathcal{M}_q$ is implemented by (11), so δ_q is a minimum of F_q on the whole fibre. Subtract $F_q(\delta_q)$ to obtain (14).

If $|A| \geq 2$, Axiom (A4) says that full revelation is strictly preferred to no information at every full-support prior. Hence $F_q(\chi_q) > 0$. Dividing by this positive number gives (15) without changing any comparison. If two affine representatives have both anchor values, positive-affine uniqueness forces the additive constant to be zero and the positive multiplier to be one. The singleton convention is immediate. $\square$

A.2.2 Stage 2: transport across priors and support faces

The remaining front-end issue is transport. The same argument handles renaming actions, deleting distinctions that the channel cannot detect, and passing to the positive support of a boundary prior. Keeping these facts in one lemma makes clear that all three are consequences of action processing, not extra invariance conventions.

For a bijection $\pi : A \rightarrow A'$, write $q\pi$ for the transported prior and $\pi_{\#}\mu$ for the posterior law obtained by transporting every atom of μ . If $B = \text{supp}(q)$, write q_B for the full-support restriction of q to B , P_B for the restricted channel, and $\pi_B\mu$ for the law obtained by restricting every posterior atom to B .

Lemma 19 (Relabelling, undetectable distinctions, and support transport). *The canonically normalized fibre values have the following properties.*

(a) Exact relabelling. *If q has full support and $\pi : A \rightarrow A'$ is a bijection, then*

$$F_{q\pi}^{A'}(\pi_{\#}\mu) = F_q^A(\mu) \quad (\mu \in \mathcal{M}_q). \quad (16)$$

(b) Undetectable distinctions. *Let $S : A \rightarrow B$ be an onto deterministic map, let q have full support, and suppose that $P : A \rightarrow \Delta(X)$ is constant on every fibre of S . Then*

$$(q, P) \sim (qS, S^qP). \quad (17)$$

In particular, if G_S reports only $S(a)$, then $(q, G_S) \sim (qS, \text{Id}_B)$.

(c) Support restriction. *For an arbitrary prior $q \in \Delta(A)$ with nonempty support $B := \text{supp}(q)$ and every channel P ,*

$$(q, P) \sim (q_B, P_B). \quad (18)$$

Consequently comparisons at q depend only on the rows indexed by B , and

$$(q, P) \succeq (q, Q) \iff (q_B, P_B) \succeq (q_B, Q_B). \quad (19)$$

Every atom of a law $\mu \in \mathcal{M}_q$ is supported on B , and the definition

$$F_q^A(\mu) := F_{q_B}^B(\pi_B\mu) \quad (20)$$

represents all continuation comparisons at the boundary prior q .

Proof. Part (a). Regard π as a deterministic action processor. Axiom (A7) gives

$$(q, P) \succeq (q\pi, P\pi^{-1}),$$

where $(P\pi^{-1})(x \mid a') = P(x \mid \pi^{-1}(a'))$. Applying the same axiom to π^{-1} gives the reverse comparison, so the two pairs are indifferent. Apply this simultaneously to channels implementing arbitrary $\mu, \nu \in \mathcal{M}_q$ and use Lemma 14 in a common four-block environment. One obtains

$$F_q^A(\mu) \geq F_q^A(\nu) \iff F_{q\pi}^{A'}(\pi_{\#}\mu) \geq F_{q\pi}^{A'}(\pi_{\#}\nu).$$

Thus $G(\mu) := F_{q\pi}^{A'}(\pi_{\#}\mu)$ is an affine representation of the same nonconstant order as F_q^A . Relabelling sends δ_q to $\delta_{q\pi}$ and χ_q to $\chi_{q\pi}$. The two anchor equalities in Lemma 18, followed by affine uniqueness, give $G = F_q^A$. If A is a singleton, both sides vanish.

Part (b). Since q has full support and S is onto, qS has full support. One direction of (17) is Axiom (A7). For the reverse direction, define the Bayesian refinement kernel

$$R(a \mid b) := \begin{cases} q(a)/(qS)(b), & S(a) = b, \\ 0, & S(a) \neq b. \end{cases}$$

Then $(qS)R = q$. Since P is constant on each fibre of S , the pushforward S^qP has at b precisely the common row of P on $S^{-1}(b)$, and pushing it back through R recovers P . Axiom (A7) applied to (qS, S^qP) and R therefore gives the reverse weak comparison. For the last assertion, G_S is S -measurable and its pushforward is Id_B .

Part (c). Choose $b_0 \in B$ and let $S : A \rightarrow B$ fix every $b \in B$ and send every $a \notin B$ to b_0 . Since q assigns zero mass outside B ,

$$qS = q_B, \quad S^qP = P_B.$$

Axiom (A7) gives $(q, P) \succeq (q_B, P_B)$. Conversely, let $\iota : B \hookrightarrow A$ be the inclusion. Then $q_B\iota = q$. The exact Bayesian completion equation pins down the processed channel on B and leaves all zero-mass rows in $A \setminus B$ arbitrary. The all-completions clause of Axiom (A7) therefore permits us to choose the completion to be the original P , yielding $(q_B, P_B) \succeq (q, P)$. This proves (18); applying it to P and Q in a common block environment proves (19). It also shows immediately that channels agreeing on B are indifferent at q .

Finally, if $\mu = \sum_k \lambda_k \delta_{s_k} \in \mathcal{M}_q$ with $\lambda_k > 0$, then for $a \notin B$,

$$0 = q(a) = \sum_k \lambda_k s_k(a).$$

Nonnegativity forces $s_k(a) = 0$ for every k . Thus every atom lies on the face $\Delta(B)$ and $\pi_B\mu \in \mathcal{M}_{q_B}$. The comparison equivalence just proved and the full-support representation on B show that (20) represents precisely the boundary continuation order. This definition is also compatible with relabelling by part (a). $\square$

At this point every later calculation may work entirely with the affine functionals F_q , their linear parts on the affine hulls of $\mathcal{M}_q$, and the exact transport identities above. No global posterior-law continuity statement and no posterior-separable integrand are needed. From now on ambient action-set superscripts are suppressed unless two action sets occur in the same formula; a boundary-prior subscript always means the representative transported from its positive support. The next two stages derive product and branch scale coherence, with Axiom (A8) and its finite-branch extension providing the essential cardinal alignment across reached branches.

A.2.3 Stage 3: products and a common comparison scale

The value F_q compares experiments only within the fibre of one prior. Embedding experiments from two priors into their product fibre will put both values on one cardinal comparison scale. We first derive the only separability fact about independent products needed for that construction. Write $P \succeq_q Q$ for $(q, P) \succeq (q, Q)$, and similarly for strict preference. For a channel $P : A \rightarrow \Delta(X)$, let $\tilde{P}$ denote its lift to $A \times B$, constant in the B -coordinate; for $R : B \rightarrow \Delta(Y)$, let $\tilde{R}$ denote the symmetric lift, constant in the A -coordinate. Posterior laws may be substituted throughout by Lemma 15; in particular, adding null records and permuting record labels has no effect on a comparison.

Lemma 20 (Cancellation of a common independent background). *Let $p \in \Delta(A)$ and $r \in \Delta(B)$ have full support. For channels P, Q on A and a channel R on B ,*

$$P \otimes R \succeq_{p \otimes r} Q \otimes R \iff P \succeq_p Q. \quad (\text{BI})$$

The symmetric assertion holds with the two coordinates interchanged. The equivalence, including its strict form, remains valid when a posterior reached in the background coordinate lies on a proper support face.

Proof. Pad P and Q by null records so that they have the same record alphabet. Regard the lift $\tilde{R}$ of R as a first-stage experiment on $A \times B$, followed on every reached branch by $\tilde{P}$, respectively $\tilde{Q}$. Up to the harmless permutation of the two record coordinates, the resulting compounds are $P \otimes R$ and $Q \otimes R$.

If y is reached under R , its posterior on $A \times B$ is

$$p \otimes r_y, \quad r_y := r_y^{r, R}.$$

The coordinate projection $A \times \text{supp}(r_y) \rightarrow A$ and the support-transport clauses of Lemmas 19 and 19 give

$$(p \otimes r_y, \tilde{P}) \sim (p, P), \quad (p \otimes r_y, \tilde{Q}) \sim (p, Q). \quad (\text{BI}')$$

Thus every reached branch has the same weak comparison as P versus Q at p . If $P \succeq_p Q$, Lemma 10 applied to these branches proves the forward implication in (BI).

For the converse, suppose that $P \not\succeq_p Q$. Completeness gives $Q \succ_p P$. By (BI'), the reversed comparison is strict on every reached branch; at least one such branch has positive mass. The strict part of Lemma 10 therefore gives $Q \otimes R \succ_{p \otimes r} P \otimes R$, contradicting $P \otimes R \succeq_{p \otimes r} Q \otimes R$. This proves the converse and also records why weak branch monotonicity alone would not suffice. Interchange the coordinates for the symmetric assertion. $\square$

For finite posterior laws define

$$\left(\sum_i m_i \delta_{u_i} \right) \otimes \left(\sum_j n_j \delta_{v_j} \right) := \sum_{i,j} m_i n_j \delta_{u_i \otimes v_j}.$$

Bayes' rule gives

$$\mu_{p \otimes r, P \otimes R} = \mu_{pP} \otimes \mu_{rR}. \quad (\text{PL})$$

Lemma 21 (Product gauge). *There are positive rescalings $G_q = c_q F_q$ of the zero-normalised affine representatives and a single number $\kappa \in \mathbb{R}$ such that*

$$G_{p \otimes r}(\mu \otimes \nu) = G_p(\mu) + G_r(\nu) + \kappa G_p(\mu) G_r(\nu) \quad (\text{QA})$$

for all full-support priors p, r and all $\mu \in \mathcal{M}_p$, $\nu \in \mathcal{M}_r$. These rescalings can be chosen exactly invariant under bijections. If a factor is a singleton, its representative is zero and the same formula holds. Finally,

$$1 + \kappa G_r(\nu) > 0 \quad (\text{POS})$$

whenever the r -fibre is nonconstant.

Proof. We give the algebra because both the common value of κ and the strict inequality (POS) are used later. Initially retain the notation F_q . For nondegenerate p, r , put

$$L_{p,r}(\mu, \nu) := F_{p \otimes r}(\mu \otimes \nu), \quad x := F_p(\mu), \quad y := F_r(\nu).$$

The map $L_{p,r}$ is separately affine. By Lemma 20, every first-coordinate slice represents the F_p -order and every second-coordinate slice represents the F_r -order. Affine-utility uniqueness, first on a slice and then as the other coordinate varies affinely, therefore gives unique constants $A_{p,r}, B_{p,r} > 0$ and $C_{p,r} \in \mathbb{R}$ such that

$$L_{p,r}(\mu, \nu) = A_{p,r}x + B_{p,r}y + C_{p,r}xy. \quad (21)$$

Here is the coefficient argument. Fixing ν first gives $L_{p,r} = \alpha(\nu)x + \gamma(\nu)$ with $\alpha(\nu) > 0$. At $\mu = \delta_p$, the function $\gamma(\nu) = L_{p,r}(\delta_p, \nu)$ is an affine representative of the F_r -order and vanishes at δ_r ; hence $\gamma(\nu) = B_{p,r}y$ for some $B_{p,r} > 0$. Choose $\bar{\mu}$ with $\bar{x} := F_p(\bar{\mu}) \neq 0$. The other slice has the form

$$L_{p,r}(\bar{\mu}, \nu) = A_{p,r}\bar{x} + D_{p,r}y,$$

where $A_{p,r} > 0$ is its value slope at $\nu = \delta_r$. Comparing this with $\alpha(\nu)\bar{x} + B_{p,r}y$ shows that $\alpha(\nu) = A_{p,r} + C_{p,r}y$, where $C_{p,r} := (D_{p,r} - B_{p,r})/\bar{x}$. This proves (21); the symmetric calculation shows that the two slice slopes obey

$$A_{p,r} + C_{p,r}y > 0, \quad B_{p,r} + C_{p,r}x > 0 \quad (22)$$

throughout the two value ranges.

Exact relabelling makes the evaluations under the two associations of a triple product equal. Comparing the coefficients of x, y, z gives

$$A_{p \otimes r, s} A_{p, r} = A_{p, r \otimes s}, \quad (23)$$

$$A_{p \otimes r, s} B_{p, r} = B_{p, r \otimes s} A_{r, s}, \quad (24)$$

$$B_{p \otimes r, s} = B_{p, r \otimes s} B_{r, s}. \quad (25)$$

The coordinate swap similarly gives

$$B_{r, p} = A_{p, r}, \quad A_{r, p} = B_{p, r}, \quad C_{r, p} = C_{p, r}. \quad (\text{SW})$$

This coefficient comparison is legitimate: the laws $(1-t)\delta_p + t\chi_p$ make x range over a nondegenerate interval, and the same holds for y and z . Let $\rho(p, r) := B_{p,r}/A_{p,r}$. Dividing (24) by (23) yields

$$\rho(p, r) = \rho(p, r \otimes s) A_{r, s}.$$

Fix a nondegenerate reference prior q_0 and set $g(r) := \rho(q_0, r)$. Then

$$A_{r, s} = \frac{g(r)}{g(r \otimes s)}. \quad (26)$$

The uniqueness of the coefficients in (21), applied after product relabellings, shows that g is invariant under bijections. Consequently the positive rescaling $G_p := g(p)F_p$ preserves exact relabelling. Equation (26) makes the new first coefficient equal to one, and (SW) makes the second coefficient equal to one as well. We henceforth use G for these gauged representatives.

Write the remaining coefficient as $\kappa_{p,r}$. Associativity of

$$x \oplus_{p,r} y := x + y + \kappa_{p,r}xy$$

and comparison of the coefficients of xy, xz, yz give

$$\kappa_{p,r} = \kappa_{p, r \otimes s}, \quad \kappa_{p \otimes r, s} = \kappa_{p, r \otimes s}, \quad \kappa_{p \otimes r, s} = \kappa_{r, s}. \quad (\text{KA})$$

(The xyz coefficient gives the product of the corresponding two sides and adds no restriction.) Taking $s = q_0$ in (KA) shows that $\kappa_{p,r} = \kappa_{r,q_0}$, so it is independent of p ; the swap in (SW) makes it symmetric, hence independent of r . Call the common value κ . After the gauge, (22) is exactly $1 + \kappa G_r(\nu) > 0$ and its symmetric counterpart.

For a singleton prior set $G \equiv 0$; the canonical bijections $A \cong A \times \{*\}$ give (QA). For a boundary prior, define G_r by exact transport from its positive-probability support, as in Lemma 19. This convention is relabelling invariant. $\square$

Corollary 22 (Exact relabelling after the product gauge). *For every bijection $\pi : A \rightarrow A'$, prior $q \in \Delta(A)$, and $\mu \in \mathcal{M}_q$,*

$$G_{q\pi}^{A'}(\pi_{\#}\mu) = G_q^A(\mu).$$

Proof. The initial equality is Lemma 19; the proof of Lemma 21 shows that its gauge factor is invariant under bijections. $\square$

Lemma 23 (Cross-prior block representation). *For arbitrary finite priors $q \in \Delta(A)$ and $r \in \Delta(B)$ and channels P, Q on the respective action sets,*

$$q^0 \succeq_{P \sqcup Q} r^1 \iff G_q(\mu_{qP}) \geq G_r(\mu_{r,Q}). \quad (\text{BC})$$

Proof. First suppose that q, r have full support. By (QA) and $G_q(\delta_q) = G_r(\delta_r) = 0$,

$$G_{q \otimes r}(\mu_{qP} \otimes \delta_r) = G_q(\mu_{qP}), \quad G_{q \otimes r}(\delta_q \otimes \mu_{r,Q}) = G_r(\mu_{r,Q}).$$

The two laws on the left lie in the same fibre and are implemented by $P \otimes U_B$ and $U_A \otimes Q$, so the affine fibre representation compares them by the displayed numbers.

It remains only to remove the uninformative product coordinates. Projection $A \times B \rightarrow A$ and Axiom (A7) give $(q \otimes r, P \otimes U_B) \succeq (q, P)$. Conversely the stochastic embedding

$$T((a, b) \mid a') = \mathbf{1}_{\{a=a'\}} r(b)$$

sends q to $q \otimes r$ and its Bayesian pushforward sends P to $P \otimes U_B$; Axiom (A7) gives the reverse comparison. Thus the two pairs are indifferent. The same argument identifies $(q \otimes r, U_A \otimes Q)$ with (r, Q) . Block coherence and transitivity now prove (BC). Arbitrary priors follow by restricting both pairs and both representatives to their supports; singleton supports use the zero convention. The singleton record tags created by these projections and embeddings are the canonical relabellings covered by the convention at the start of the appendix. $\square$

A.2.4 Stage 4: sequential decomposition and coherent face scales

Changing a continuation on one reached branch changes the posterior law by a signed measure with zero total mass and zero barycentre. The following space is exactly the space of such feasible directions. For a nonempty finite set A , let

$$W_A := \left\{ \eta : \begin{array}{l} \eta \text{ is a finitely supported signed measure on } \Delta(A), \\ \eta(\Delta(A)) = 0, \quad \int p \, d\eta(p) = 0 \end{array} \right\}.$$

If $\iota : B \hookrightarrow A$ is an injection, let $i_\iota : W_B \rightarrow W_A$ push every posterior forward along ι ; for a literal subset $B \subseteq A$, write i_B for the inclusion case. Let L_q^A be the linear part of the affine functional G_q on the affine hull of $\mathcal{M}_q$.

Lemma 24 (Branch aggregation). *Let q have full support on A . For every reached posterior r with $B := \text{supp}(r)$ and $|B| \geq 2$, there is a unique $\beta_A(q, r) > 0$ satisfying*

$$L_q^A \circ i_B = \beta_A(q, r) L_{r_B}^B \quad \text{on } W_B. \quad (27)$$

It depends only on q, r and the face inclusion, not on the experiment that reaches r . If a first-stage channel $P_1 : A \rightarrow \Delta(\{1, \dots, k\})$ has branch masses m_i and posteriors r_i , then every list of continuation channels $Q_1, \dots, Q_k$ satisfies

$$G_q(\mu_{q, P_1 \triangleright (Q_1, \dots, Q_k)}) = G_q(\mu_{q, P_1}) + \sum_{i: m_i > 0} m_i \beta_A(q, r_i) G_{r_i}(\mu_{r_i, Q_i}). \quad (\text{BA})$$

For singleton r_i , the continuation value is zero and the corresponding coefficient may be fixed by any positive convention.

Proof. We first record the common tangent space. If q has full support, then

$$\text{span}\{\mu - \nu : \mu, \nu \in \mathcal{M}_q\} = W_A. \quad (28)$$

Only the reverse inclusion needs proof. For $0 \neq \eta \in W_A$, write its Jordan decomposition as $\eta = \eta^+ - \eta^-$, where both positive parts have the same mass $c > 0$ and, after division by c , the same barycentre b . Choose $\varepsilon > 0$ small enough that $\tau = (q - \varepsilon b)/(1 - \varepsilon)$ belongs to $\Delta(A)$. Then

$$\mu^\pm := \frac{\varepsilon}{c} \eta^\pm + (1 - \varepsilon) \delta_\tau \in \mathcal{M}_q, \quad \eta = \frac{c}{\varepsilon} (\mu^+ - \mu^-),$$

which proves (28). The same argument intrinsically on a face gives the space W_B .

Fix a nondegenerate r supported on B . It is reachable from q : choose

$$0 < \varepsilon < \min \left\{ 1, \min_{b \in B} \frac{q(b)}{r(b)} \right\}$$

and use a two-record experiment whose first record has conditional probability $\varepsilon r(a)/q(a)$. That record has mass ε and posterior r .

Implement $\nu, \nu' \in \mathcal{M}_{r_B}$ by channels on B , extend their off-support rows arbitrarily to A , and pad their record alphabets if necessary. Vary only the continuation on the branch reaching r from ν to ν' . The aggregate posterior law changes by $\varepsilon i_B(\nu - \nu')$. Corollary 11, support transport, and strictness give all three signs:

$$\begin{aligned} L_{r_B}^B(\nu - \nu') > 0 &\implies L_q^A i_B(\nu - \nu') > 0, \\ L_{r_B}^B(\nu - \nu') < 0 &\implies L_q^A i_B(\nu - \nu') < 0, \\ L_{r_B}^B(\nu - \nu') = 0 &\implies L_q^A i_B(\nu - \nu') = 0. \end{aligned}$$

The second line is the first applied after swapping the continuations. For the third, indifference supplies both weak branch comparisons, and Corollary 11 supplies both aggregate inequalities. By (28), the sign agreement extends by positive scaling to all of W_B . Positive trace orientation makes

$$L_{r_B}^B(\chi_{r_B} - \delta_{r_B}) = G_{r_B}(\chi_{r_B}) > 0,$$

so $L_{r_B}^B \neq 0$. Two nonzero linear functionals with the same positive half-space have the same kernel and are positive scalar multiples; this gives the unique coefficient in (27). Since both linear functionals in that equation are fixed before a reaching experiment is chosen, the coefficient is path-independent.

For the aggregation formula, put $B_i := \text{supp}(r_i)$ and

$$\nu_i := \pi_{B_i} \mu_{r_i, Q_i} \in \mathcal{M}_{(r_i)_{B_i}}.$$

The compound law differs from the first-stage law by

$$\mu_{q, P_1 \triangleright (Q_1, \dots, Q_k)} - \mu_{q, P_1} = \sum_{i: m_i > 0} m_i i_{B_i}(\nu_i - \delta_{(r_i)_{B_i}}).$$

Apply L_q^A , use (27), and use $G_{r_i}(\delta_{r_i}) = 0$. Singleton summands vanish. This is (BA). $\square$

For an injection $\iota : B \hookrightarrow A$ and full-support $r \in \Delta(B)$, define

$$\beta_\iota(q, r) := \beta_A(q, \iota_\# r).$$

The image posterior $\iota_\# r$ is reachable by the two-record construction in the proof, and its support inclusion is precisely i_ι . Thus

$$L_q^A i_\iota = \beta_\iota(q, r) L_r^B \quad \text{on } W_B.$$

Lemma 25 (Cocycle, face coherence, and the chain gauge). *The coefficients of Lemma 24 admit positive scales a_q^A which are invariant under relabelling and satisfy*

$$\beta_\iota(q, r) = \frac{a_q^A}{a_r^B} \tag{FS}$$

whenever $\iota : B \hookrightarrow A$, both sets are nondegenerate, q and r have full support on their respective sets, and $\beta_\iota(q, r)$ is the coefficient for reaching the face $\iota(B)$. Hence the representatives

$$V_q := \frac{G_q}{a_q}$$

satisfy, for every $P_1 : A \rightarrow \Delta(\{1, \dots, k\})$ with branch masses m_i and posteriors r_i , and every list of continuation channels $Q_1, \dots, Q_k$, the exact chain rule

$$V_q(\mu_{q, P_1 \triangleright (Q_1, \dots, Q_k)}) = V_q(\mu_{q, P_1}) + \sum_{i: m_i > 0} m_i V_{r_i}(\mu_{r_i, Q_i}). \tag{CR}$$

Singleton continuation terms are zero.

Proof. There are two normalization tasks: first solve the branch cocycle within each fixed action-set dimension, then remove the residual scalar attached to embedding an m -point face in an n -point simplex.

We first prove the cocycle before choosing scales. For nested injections $C \xrightarrow{j} B \xrightarrow{\iota} A$ and full-support priors q, r, s on A, B, C , respectively, the defining identities on W_C are

$$L_q^A i_{\iota \circ j} = \beta_\iota(q, r) L_r^B i_j = \beta_\iota(q, r) \beta_j(r, s) L_s^C.$$

The direct coefficient from q to s is unique because $L_s^C \neq 0$ by Axiom (A4). Therefore

$$\beta_{\iota \circ j}(q, s) = \beta_\iota(q, r) \beta_j(r, s). \tag{29}$$

This single identity covers full-support paths, full-to-face paths, and nested support faces.

For a fixed nondegenerate A , apply (29) to identity embeddings. Fix a full-support basepoint q_A and put

$$a_q^A := \beta_{\text{id}_A}(q, q_A).$$

Then

$$\beta_{\text{id}_A}(q, r) = \frac{a_q^A}{a_r^A}. \tag{WS}$$

Choosing q_A to be uniform and using exact relabelling of G makes these initial scales invariant under bijections. They remain free up to one positive multiplier for each cardinality.

For an injection $\iota : B \hookrightarrow A$, define its residual defect by

$$K(\iota) := \beta_\iota(q, r) \frac{a_r^B}{a_q^A}.$$

Equation (WS) shows that this number is independent of q and r : changing either prior merely inserts and cancels the corresponding within-set coefficient. Equation (29) gives

$$K(\iota \circ j) = K(\iota)K(j).$$

Exact relabelling implies that $K(\iota)$ depends only on $n = |A|$ and $m = |B|$; write it $K_{n,m}$. Thus

$$K_{n,\ell} = K_{n,m}K_{m,\ell}, \quad K_{n,n} = 1.$$

Set $t_n := K_{n,2}$ and $t_2 := 1$. Then $K_{n,m} = t_n/t_m$. Replacing every a_q^A on an n -point set by $t_n a_q^A$ changes the defect to $K_{n,m} t_m/t_n = 1$, while preserving (WS) and relabel invariance. This proves (FS).

For a boundary prior, a_r means the scale on its support; the preceding construction says precisely that this agrees with the scale obtained by reaching that support as a face. Divide (BA) by a_q and use (FS). Every nondegenerate branch coefficient becomes

$$\frac{1}{a_q} \beta_A(q, r_i) G_{r_i} = \frac{1}{a_{r_i}} G_{r_i} = V_{r_i}.$$

Singleton values vanish, so their scales and coefficients may be fixed arbitrarily. This proves (CR). $\square$

A.2.5 Stage 5: compatibility and entropy identification

The product and sequential evaluations of full revelation now meet. Their compatibility first removes the provisional product interaction and then leaves a strongly additive uncertainty functional.

Lemma 26 (Interaction collapse and universal chain scale). *The coefficient κ in (QA) is zero. Moreover there is a single $C > 0$ such that $a_q = C$ for every nondegenerate finite prior, read on its support. The same value may be assigned to singleton priors. Consequently the final representatives $V_q = G_q/C$ satisfy both*

$$V_{p \otimes r}(\mu \otimes \nu) = V_p(\mu) + V_r(\nu), \quad (\text{PA})$$

$$V_q(\mu_{q, P_1 \triangleright (Q_1, \dots, Q_k)}) = V_q(\mu_{q P_1}) + \sum_{i: m_i > 0} m_i V_{r_i}(\mu_{r_i, Q_i}). \quad (\text{CR}')$$

for every $P_1 : A \rightarrow \Delta(\{1, \dots, k\})$ with branch masses m_i and posteriors r_i , and every list of continuation channels $Q_1, \dots, Q_k$. The cross-prior bridge of Lemma 23 is valid with V in place of G .

Proof. In this proof only, put

$$H_G(q) := G_q(\chi_q), \quad Z(q) := 1 + \kappa H_G(q).$$

For nondegenerate q , Axiom (A4) and zero normalisation give $H_G(q) > 0$, while the strict slice-slope (POS) gives

$$Z(q) > 0. \quad (30)$$

For a singleton, $H_G = 0$ and $Z = 1$.

Product revelation versus sequential revelation. For nondegenerate full-support p, r , product quasi-additivity gives

$$H_G(p \otimes r) = H_G(p) + H_G(r) + \kappa H_G(p) H_G(r). \quad (\text{QH})$$

Compute the same full revelation by first revealing the A -coordinate and then the B -coordinate. The first-stage law is $\chi_p \otimes \delta_r$ and has value $H_G(p)$ by (QA). After action a is revealed, the

posterior is $e_a \otimes r$ on the face $\{a\} \times B$; support restriction and relabelling identify the continuation value with $H_G(r)$. Every such branch has coefficient $a_{p \otimes r}/a_r$ by (FS). Since the branch weights sum to one, Lemma 24 gives

$$H_G(p \otimes r) = H_G(p) + \frac{a_{p \otimes r}}{a_r} H_G(r). \quad (\text{SA})$$

Comparing (QH) and (SA), and using $H_G(r) > 0$, yields

$$\frac{a_{p \otimes r}}{a_r} = Z(p). \quad (\text{R1})$$

Revealing B first gives symmetrically

$$\frac{a_{p \otimes r}}{a_p} = Z(r). \quad (\text{R2})$$

Thus $a_r Z(p) = a_p Z(r)$ for every pair of nondegenerate priors, even on different finite sets. By (30), there is a universal $C > 0$ such that

$$a_q = CZ(q). \quad (\text{AS})$$

Assign $a = C$ on singleton supports.

If $\kappa = 0$, then $Z \equiv 1$ and the conclusion already follows. It remains to rule out $\kappa \neq 0$.

The grouping weight. Let q have full support and let $\mathcal{P} = \{A_1, \dots, A_m\}$ partition its action set into nonempty cells. Put

$$p_i = q(A_i), \quad q^i = q(\cdot \mid A_i), \quad p = (p_1, \dots, p_m).$$

If $C_{\mathcal{P}}$ reports only the cell, undetectable-distinctions neutrality gives

$$(q, C_{\mathcal{P}}) \sim (p, \text{Id}_{\{1, \dots, m\}}).$$

The cross-prior bridge therefore identifies the coarse value as

$$G_q(\mu_{q, C_{\mathcal{P}}}) = H_G(p). \quad (\text{BG})$$

Append full revelation within each cell. The final law is χ_q , and the branch formula together with (AS) gives

$$H_G(q) = H_G(p) + Z(q) \sum_i p_i \frac{H_G(q^i)}{Z(q^i)}. \quad (\text{GR})$$

Here singleton cells contribute zero.

Set $w(q) := 1/Z(q) > 0$. Because $\kappa \neq 0$ in the present paragraph, $\kappa H_G(x) = Z(x) - 1$. Substitution into (GR) gives

$$Z(q) - 1 = Z(p) - 1 + Z(q) \sum_i p_i \left(1 - \frac{1}{Z(q^i)}\right).$$

Cancelling and inverting yields the positive grouping-weight equation

$$w(q) = w(p) \sum_i p_i w(q^i). \quad (\text{W})$$

All distributions in what follows are read on their positive-probability supports. For $u \otimes v$, partition by the first coordinate: the coarse distribution is u and every conditional distribution is v . Thus (W) gives

$$w(u \otimes v) = w(u)w(v). \quad (\text{M})$$

Two groupings force the weight to be constant. Take arbitrary finite probability vectors u, v and write $x = w(u)$ and $y = w(v)$. Let T be the law of a triple (ℓ, z_1, z_2) : ℓ is uniform on $\{0, 1\}$, and conditional on $\ell = 0$ the pair (z_1, z_2) has law $u \otimes u$, while conditional on $\ell = 1$ it has law $v \otimes v$. In tagged notation,

$$T := \frac{1}{2}(u \otimes u)^0 \sqcup \frac{1}{2}(v \otimes v)^1.$$

Grouping first by ℓ and using (M) gives

$$w(T) = w(\frac{1}{2}, \frac{1}{2}) \frac{x^2 + y^2}{2}. \quad (\text{E1})$$

Instead group first by (ℓ, z_1) . That marginal is $S = \frac{1}{2}u^0 \sqcup \frac{1}{2}v^1$, so

$$w(S) = w(\frac{1}{2}, \frac{1}{2}) \frac{x + y}{2}.$$

The conditional law of z_2 is u when $\ell = 0$ and v when $\ell = 1$. A second use of (W) therefore gives

$$w(T) = w(S) \frac{x + y}{2} = w(\frac{1}{2}, \frac{1}{2}) \left(\frac{x + y}{2} \right)^2. \quad (\text{E2})$$

The common leading factor is positive. Comparing (E1) and (E2) yields $(x - y)^2 = 0$. Thus w is constant on all finite probability vectors. Its singleton value is one, so $w \equiv 1$.

It follows that $Z \equiv 1$, hence $\kappa H_G(q) = 0$ for every q . Axiom (A4) supplies a nondegenerate q with $H_G(q) > 0$, so $\kappa = 0$, contradicting the temporary alternative. Equation (AS) now gives $a_q = C$ universally. Dividing the product and branch formulas by C proves (PA) and (CR'). Since the division is by one common positive number, it also preserves the cross-prior bridge. $\square$

Entropy identification.

Lemma 27 (Entropy reduction and Faddeev identification). *Define the value of full revelation by*

$$\mathcal{H}(q) := V_q(\chi_q),$$

so $\mathcal{H}(q) = H_G(q)/C$, with boundary priors read on their support. Then every finite channel $P : A \rightarrow \Delta(X)$ satisfies

$$V_q(\mu_{qP}) = \mathcal{H}(q) - \sum_{x: m_{qP}(x) > 0} m_{qP}(x) \mathcal{H}(r_x^{q,P}). \quad (\text{ER})$$

Moreover,

$$\mathcal{H}(q) = \alpha H(q)$$

for a single $\alpha > 0$. Consequently

$$V_q(\mu_{qP}) = \alpha I_{qP}(A; X). \quad (\text{MI})$$

Proof. Entropy reduction. Assume first that q has full support. Append full revelation after P . The final posterior law is χ_q , while the continuation after record x has full-revelation law χ_{r_x} on the support of r_x . The exact chain rule (CR') therefore gives

$$\mathcal{H}(q) = V_q(\mu_{qP}) + \sum_{x: m_{qP}(x) > 0} m_{qP}(x) \mathcal{H}(r_x),$$

which is (ER). A boundary prior is reduced to its support; null action rows and null records affect neither side.

Strong additivity. Let $\mathcal{P} = \{A_1, \dots, A_m\}$ be a partition of the support of q , and put

$$p_i = q(A_i), \quad q^i = q(\cdot \mid A_i), \quad p = (p_1, \dots, p_m).$$

The channel $C_{\mathcal{P}}$ which reports the cell is neutral to the corresponding full revelation on the cell index:

$$(q, C_{\mathcal{P}}) \sim (p, \text{Id}_{\{1, \dots, m\}})$$

by Lemma 19. The common-scale cross-prior bridge from Lemma 26 therefore gives

$$V_q(\mu_{q, C_{\mathcal{P}}}) = \mathcal{H}(p).$$

Applying (ER) to $C_{\mathcal{P}}$ yields

$$\mathcal{H}(q) = \mathcal{H}(p) + \sum_i p_i \mathcal{H}(q^i). \quad (\text{FA})$$

Thus $\mathcal{H}$ is strongly additive. Relabelling invariance of V makes $\mathcal{H}$ symmetric; support restriction makes it expansive under adding zero-probability states; and $\mathcal{H}(\delta_a) = 0$. Record processing makes every experiment weakly preferable to its no-information garbling, so $V_q(\mu) \geq 0$ on every fibre and hence $\mathcal{H}(q) \geq 0$. Axiom (A4) gives strict positivity at every nondegenerate full-support prior. Although (FA) was derived by partitioning the positive support, it also holds with zero outer weights: delete the null cells by support restriction, apply the displayed recursion, and reinsert them by expansibility.

Continuity: the binary case. Write $h(t) := \mathcal{H}(t, 1 - t)$. We derive, rather than assume, its continuity from Axiom (A2). Fix $c \geq 0$. Choose a full-support binary prior θ with $\mathcal{H}(\theta) > 0$, as supplied by Axiom (A4). Choose $n \geq 1$ and $\lambda \in [0, 1]$ so that

$$c = \lambda n \mathcal{H}(\theta).$$

For $\theta_n := \theta^{\otimes n}$, product additivity (PA) gives $\mathcal{H}(\theta_n) = n \mathcal{H}(\theta)$. Let $E_{n, \lambda}$ reveal the n -bit action fully with probability λ and report a common erasure symbol otherwise. Its posterior law is $\lambda \chi_{\theta_n} + (1 - \lambda) \delta_{\theta_n}$, so affinity gives

$$V_{\theta_n}(\mu_{\theta_n, E_{n, \lambda}}) = c.$$

Now fix the block environment

$$K_c := \text{Id}_{\{0, 1\}} \sqcup E_{n, \lambda}.$$

Let q_t^0 put the binary prior $(t, 1 - t)$ on its first block and let θ_n^1 put θ_n on its second block. The cross-prior bridge says

$$\begin{aligned} h(t) \geq c &\iff q_t^0 \succeq_{K_c} \theta_n^1, \\ h(t) \leq c &\iff \theta_n^1 \succeq_{K_c} q_t^0. \end{aligned}$$

The environment is fixed and $t \mapsto q_t^0$ is continuous, so Axiom (A2) makes both level sets closed. When $c < 0$, the superlevel set is all of $[0, 1]$ and the sublevel set is empty because $h \geq 0$. Thus every upper and lower level set of h is closed, and h is continuous.

Continuity on every finite simplex. Proceed by induction on the number of coordinates. For $q = (q_1, (1 - q_1)\bar{q})$ with $q_1 < 1$, recursion (FA) for the partition $\{\{1\}, \{2, \dots, n\}\}$ gives

$$\mathcal{H}(q) = h(q_1) + (1 - q_1) \mathcal{H}(\bar{q}). \quad (31)$$

The right side is continuous away from $q_1 = 1$ by binary continuity and the induction hypothesis. At $q_1 = 0$, support restriction and $h(0) = 0$ give the same formula on the face. At $q_1 = 1$, the induction hypothesis makes $\mathcal{H}$ bounded on the compact $(n - 1)$ -simplex, so the second term in

(31) tends to zero and the first tends to $h(1) = 0$. Relabelling handles the other faces. Hence $\mathcal{H}$ is continuous on each closed finite simplex.

We have now verified all hypotheses of the strong-additivity form of Faddeev's theorem: continuity on each finite simplex, invariance under bijections, expansibility, vanishing on point masses, and (FA). The finite entropy characterisation [7, Theorem 6] therefore gives

$$\mathcal{H}(q) = \alpha H(q)$$

for some $\alpha \geq 0$. Axiom (A4) forces $\alpha > 0$. Substitution in (ER) gives

$$V_q(\mu_{qP}) = \alpha \left(H(q) - \sum_x m_{qP}(x) H(r_x^{q,P}) \right) = \alpha I_{qP}(A; X),$$

where the sum is over positive-mass records. This proves (MI). $\square$

Completion of Proposition 13. After Lemma 26, the cross-prior bridge is represented by V . Lemma 27 identifies this value as αI with one $\alpha > 0$. Hence, for arbitrary pairs (q, P) and (r, Q) ,

$$q^0 \succeq_{P \sqcup Q} r^1 \iff I_{qP} \geq I_{r,Q}, \quad (\text{BMI})$$

including boundary priors by support restriction. For one fixed channel, the duplication clause of Axiom (A5) identifies $q \succeq_P q'$ with the left side of (BMI) for $Q = P$, proving the first assertion. For the block clause, Axiom (A5) deletes all irrelevant blocks and (BMI) compares the two remaining pairs. A lottery supported on block i has a constant block tag, so

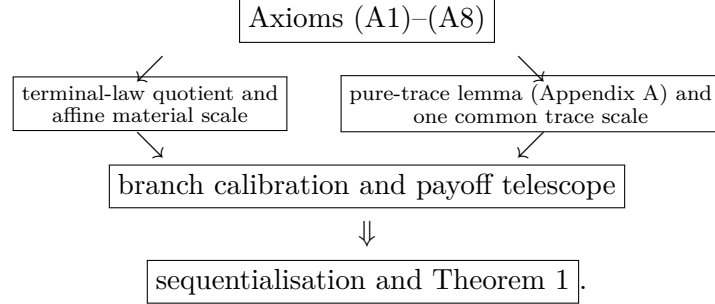
$$I_{q_i^i, \bigsqcup_{k \in J} P_k} = I_{q_i P_i},$$

and similarly for block j . This is exactly the stated common-scale block comparison. $\square$

B Completion of the representation theorem

Appendix A proves the one substantial auxiliary input: the pure-trace lemma on the constant-payoff domain. This appendix combines that result with the affine material scale, calibrates deterministic payoff branches, and completes Theorem 1. We also use the ordinary mixture-space theorem: a complete and transitive mixture order satisfying independence and segment calibration has an affine representation.

The logical order is therefore



The left branch identifies the value of terminal payoffs and fixes a common material scale. The right branch is proved once on the constant-payoff domain. They meet in the deterministic branch calculation, which aligns material increments with the trace scale and telescopes across all reached branches.

B.1 Preliminaries

We first dispose of three pieces of bookkeeping. They will be used without further comment.

Recall that $(q, K) \succeq (p, L)$ denotes the paper's canonical two-block comparison. For channels E, G at a common prior q , write $E \succeq_q G$ for $(q, E) \succeq (q, G)$. A *finite experiment at q* is a finite payoff-record channel paired with that prior; its marked terminal law is denoted by η_E .

Lemma 28 (Blocks, supports, and dummies). *Suppose Axioms (A1), (A2), and (A5)–(A7) hold.*

- (a) *After identifying alternatives related by invertible action or record relabellings, the derived comparison of pairs is complete and transitive. If either member of a pair comparison is replaced by an indifferent alternative, the comparison is unchanged.*
- (b) *If $S = \text{supp } q$, then (q, K) is indifferent to the restriction of (q, K) to S .*
- (c) *If $s \in \Delta(B)$ and $K^\uparrow(o, r \mid a, b) := K(o, r \mid a)$, then*

$$(q \otimes s, K^\uparrow) \sim (q, K). \quad (32)$$

Proof. Place any finite collection of alternatives in one finite block environment. Axiom (A5) removes unused blocks, and Axiom (A1) supplies completeness and transitivity there. Invertible action and record relabellings are neutral by applying Axioms (A6) and (A7) in both directions. In particular, swapping both tags identifies the two orientations of a two-block comparison. This proves part (a); a three-block environment gives transitivity and a four-block environment gives replacement.

For part (b), process an action in S by its inclusion in A . In the other direction, output $a \in S$ when the realised action is a , and choose an arbitrary element of S off the support. Both processors satisfy (1); rows attached to null processed actions may be completed arbitrarily. Axiom (A7) gives the two comparisons.

For part (c), first project (a, b) to a . Conversely, given a , draw b independently from s and output (a, b) . Again the two joint-law identities in (1) are exact, so Axiom (A7) applies in both directions. Part (a) completes both arguments. $\square$

Fix for the moment a full-support prior $q \in \Delta(A)$. For a joint channel K , write

$$p_K(o, r) := \sum_a q(a) K(o, r | a), \quad \pi_{or}^K(a) := \frac{q(a) K(o, r | a)}{p_K(o, r)}$$

when $p_K(o, r) > 0$, and define its *marked terminal law*

$$\eta_K := \sum_{(o, r): p_K(o, r) > 0} p_K(o, r) \delta_{(o, \pi_{or}^K)}. \quad (33)$$

It records the terminal payoff and the posterior about the realised action. Write $I_q(E)$ for the mutual information between the action and the complete visible output of E under q . When the output is a first-stage record Y , we also write this quantity as $I_q(A; Y)$.

Lemma 29 (Terminal laws and affine fibres). *Under Axioms (A1), (A2), and (A5)–(A8), the following statements hold.*

- (a) *If q has full support and $\eta_K = \eta_L$, then $(q, K) \sim (q, L)$.*
- (b) *At every full-support q , the quotient of finite experiments by their marked terminal laws has an affine representation W_q .*

Proof. We give the finite Blackwell argument behind part (a). For a payoff o and posterior π , let

$$M_K(o, \pi) := \sum_{\substack{r: p_K(o, r) > 0 \\ \pi_{or}^K = \pi}} p_K(o, r).$$

Equality of marked laws gives $M_K(o, \pi) = M_L(o, \pi) =: M(o, \pi)$. For $p_K(o, r) > 0$, draw a record s of L according to

$$T(s | o, r) := \mathbf{1}_{\{p_L(o, s) > 0, \pi_{os}^L = \pi_{or}^K\}} \frac{p_L(o, s)}{M(o, \pi_{or}^K)}. \quad (34)$$

Choose an arbitrary probability row when $p_K(o, r) = 0$. This is a payoff-preserving record processor. Bayes' formula gives

$$q(a) K(o, r | a) = p_K(o, r) \pi_{or}^K(a).$$

For a reached target record s , with posterior $\pi = \pi_{os}^L$, it follows that

$$\sum_r K(o, r | a) T(s | o, r) = \frac{p_L(o, s)}{M(o, \pi)} \frac{\pi(a)}{q(a)} M(o, \pi) = L(o, s | a).$$

If s is unreached, full support makes both sides zero. Thus postprocessing K by T gives L . The symmetric construction gives a processor from L to K . Axiom (A6) and Lemma 28(a) give indifference. For a boundary prior, Lemma 28(b) first deletes the null actions.

For part (b), take the quotient of actual finite marked experiments by equality of (33). If E and G are two such experiments, their public mixture first draws a coin independent of the action, runs the selected experiment, and retains the coin in the record. Its marked law is

$$\eta_{tE \oplus (1-t)G} = t\eta_E + (1-t)\eta_G. \quad (35)$$

Axiom (A8) gives, for $0 < t < 1$ and any third experiment L ,

$$E \succeq_q L \iff tE \oplus (1-t)G \succeq_q tL \oplus (1-t)G. \quad (36)$$

This is the action-independent special case of the axiom: the public coin is the binary first-stage record and G is the common continuation. If record alphabets differ, put them in a common disjoint union before invoking the axiom; part (a) identifies the result with (35).

It remains only to check the Archimedean step. A closed segment between two experiments, and any fixed comparator, can be realised on one fixed union of their finite record alphabets. Its channel entries vary continuously with $t \in [0, 1]$. Axiom (A2) therefore makes both contour sets along the segment closed. Completeness says that they cover $[0, 1]$; when a target lies between the two endpoints, the sets contain opposite endpoints and connectedness makes them intersect. Thus every such target is indifferent to a point of the segment. The mixture-space theorem now supplies an affine representative W_q . This quotient is closed under every mixture and embedding used below. $\square$

B.2 The two common scales

Lemma 30 (Material utility and a common affine scale). *Under Axioms (A1)–(A3) and (A5)–(A8), there are outcomes o^+, o^- and a nonconstant function $u : O \rightarrow \mathbb{R}$ such that*

$$u(o^+) = 1, \quad u(o^-) = 0.$$

The payoff-lottery order is represented by $\sum_o \ell(o)u(o)$. The representatives in Lemma 29 may be normalised so that

$$W_q(\ell) = \sum_o \ell(o)u(o). \quad (37)$$

With this normalisation, adjoining a full-support independent dummy to a full-support prior preserves value, and for full-support q, q' ,

$$(q, E) \succeq (q', G) \iff W_q(E) \geq W_{q'}(G). \quad (38)$$

Proof. On a singleton action set, terminal-law sufficiency identifies an experiment with its payoff lottery. Public-mixture independence and continuity, proved as in Lemma 29, give an affine utility on $\Delta(O)$. By Lemma 12, Axiom (A3) supplies $o^+ \succ o^-$. Normalise their values to one and zero and let $u(o)$ be the value of the degenerate lottery at o . Affinity on the finite simplex gives expected utility.

An action processor may ignore its input and draw an arbitrary target prior. Axiom (A7), applied in both directions, therefore makes this payoff lottery order independent of the prior and action alphabet. Normalise each W_q at the same two deterministic payoffs. The payoff-lottery embedding is affine and order-reflecting, so positive-affine uniqueness gives (37).

Dummy lifting between full-support fibres is likewise an affine order embedding. Positive-affine uniqueness initially permits

$$W_{q \otimes s}(E^\dagger) = cW_q(E) + d, \quad c > 0.$$

The two payoff anchors have values zero and one on both sides; hence $c = 1$ and $d = 0$. To compare E at q with G at q' , lift each to the common product prior $q \otimes q'$, using the other prior as the dummy. Equation (32), the replacement part of Lemma 28, and the common product representative give (38). $\square$

Let o_* be the outcome fixed in Appendix A to witness Axiom (A4). We now use Proposition 13 on the constant- o_* lift

$$K_P(o, r \mid a) = \mathbf{1}_{\{o=o_*\}}P(r \mid a).$$

This lift commutes with block formation, record processing, action processing, limits on each finite simplex, and compounding. Thus Axiom (A1) gives the pure weak order, while Lemma 12 turns Axiom (A4) into its positive orientation; Axiom (A2) gives continuity; Axiom (A5) gives block coherence; Axioms (A6) and (A7) give the two data processing conditions, including every null-row completion; and Lemma 10 gives finite-branch continuation. These are precisely the hypotheses of Proposition 13. Its conclusion is that every within-channel and block-supported comparison between such lifts is ordered by $I_{qP}(A; R)$.

Lemma 31 (The trace scale). *Under Axioms (A1)–(A8), there is one number $\lambda > 0$, independent of the prior and all finite alphabets, such that every full-support, nontrivial action fibre and every experiment E that pays o_* surely satisfy*

$$W_q(E) = u(o_*) + \lambda I_q(E). \quad (39)$$

Proof. Fix a nontrivial full-support q . On the constant- o_* face, W_q is affine by Lemma 29. For any experiment E , the finite entropy identity

$$I_q(E) = H(q) - \sum_{(o,\pi) \in \text{supp } \eta_E} \eta_E(o, \pi) H(\pi). \quad (40)$$

shows that mutual information depends only on the marked terminal law and is affine under public mixtures. Proposition 13 says that W_q and I_q represent the same nonconstant order on the constant- o_* face. Positive-affine uniqueness therefore gives

$$W_q(K_P) = \lambda_q I_{qP} + b_q, \quad \lambda_q > 0.$$

The uninformative experiment has mutual information zero and, by (37), value $u(o_*)$. Thus $b_q = u(o_*)$.

The coefficient does not depend on q . Let q and q' be nontrivial full-support priors. Lift full revelation at q to $q \otimes q'$ by adding an independent dummy coordinate. Normalised value and mutual information are both unchanged. Since the common information value is $H(q) > 0$, cancellation gives $\lambda_{q \otimes q'} = \lambda_q$. Lifting full revelation from the other coordinate gives $\lambda_{q \otimes q'} = \lambda_{q'}$. Hence all such coefficients agree. Equivalently, fix the uniform prior on a two-point dummy alphabet and call its coefficient λ . Axiom (A4) fixes the strict orientation, so $\lambda > 0$.

Finally, any experiment that pays o_* surely is a pure record experiment after the constant payoff coordinate is deleted; restoring that coordinate changes neither its marked terminal law nor its mutual information. This proves (39). $\square$

B.3 The branch calculation

Let q be full support on a nontrivial A and let $P : A \rightarrow \Delta(Y)$ be a first-stage record channel. Put

$$m_y := \sum_a q(a) P(y | a).$$

For a deterministic payoff profile $g : Y \rightarrow O$, let $C(P, g)$ denote the compound that first runs P and then pays $g(y)$ with no further record. Write g_y^o for the profile that pays o at y and o_* elsewhere.

Lemma 32 (The reached-branch payoff increment). *For every $y \in Y$ and $o \in O$,*

$$W_q(C(P, g_y^o)) - W_q(C(P, o_*)) = m_y [u(o) - u(o_*)]. \quad (41)$$

Proof. If $m_y = 0$, changing the payoff on branch y does not change the marked terminal law, so both sides are zero. Suppose $m_y > 0$, and put

$$r_y(a) := \frac{q(a) P(y | a)}{m_y}, \quad S_y := \text{supp } r_y.$$

Let $\bar{r}_y$ be the full-support restriction of r_y to S_y . For a marked continuation E on S_y , extend E to A through a support projection, insert it at branch y , and put the payoff o_* for sure at every other branch. Denote the resulting experiment by $J_y(E)$. If $a \notin S_y$, then $r_y(a) = 0$; since $q(a) > 0$ and $m_y > 0$, the definition of r_y gives $P(y | a) = 0$. Hence the continuation rows chosen

outside S_y are never used on branch y . Support restriction identifies their local versions as well, so the extension is independent of the chosen projection.

Support restriction and Corollary 11 give

$$E \succeq_{\bar{r}_y} G \iff J_y(E) \succeq_q J_y(G).$$

Different continuation record alphabets are first put in a common disjoint union. Moreover, J_y commutes with public mixtures at the level of marked terminal laws. Thus affine uniqueness gives constants $c_y > 0, d_y$ such that

$$W_q(J_y(E)) = c_y W_{\bar{r}_y}(E) + d_y. \quad (42)$$

Suppose first that S_y has at least two elements. At payoff o_* , compare full revelation, E^{id} , with an uninformative record, E^0 . Lemma 31 gives the local difference

$$W_{\bar{r}_y}(E^{\text{id}}) - W_{\bar{r}_y}(E^0) = \lambda H(\bar{r}_y).$$

The finite mutual-information chain rule for insertion is

$$I_q(J_y(E)) = I_q(A; Y) + m_y I_{\bar{r}_y}(E). \quad (43)$$

Support extension does not change the last term. Both inserted experiments pay o_* surely, so Lemma 31 and (43) make their aggregate value difference $\lambda m_y H(\bar{r}_y)$. Equation (42) makes the same difference $c_y \lambda H(\bar{r}_y)$. Since $\lambda > 0$ and a nontrivial full-support distribution has positive entropy, $c_y = m_y$.

Locally, deterministic payoffs o and o_* have values $u(o)$ and $u(o_*)$ by (37). Subtracting their two instances of (42), and identifying the resulting insertions with the two deterministic compounds in the statement, proves (41).

If S_y is a singleton, append an independent full-support two-point dummy action to the first-stage action. The reached posterior becomes the product of r_y and the dummy prior and hence has nontrivial support. The branch mass is still m_y , and Lemma 30 says that both compound values are unchanged. Apply the preceding argument in the product problem and remove the dummy. $\square$

Lemma 33 (Deterministic branch assembly). *For every deterministic payoff profile $g : Y \rightarrow O$,*

$$W_q(C(P, g)) = \lambda I_q(A; Y) + \sum_y m_y u(g(y)). \quad (44)$$

Proof. First observe that the increment from changing one branch is independent of the payoffs on the other branches. Indeed, for any two backgrounds g, h , the two crossed public half-mixtures

$$\frac{1}{2}C(P, g[y \mapsto o]) \oplus \frac{1}{2}C(P, h[y \mapsto o_*])$$

and

$$\frac{1}{2}C(P, g[y \mapsto o_*]) \oplus \frac{1}{2}C(P, h[y \mapsto o])$$

have the same marked terminal law. Away from y they contain the same two payoff-posterior atoms; at y they contain o and o_* with the same mass $m_y/2$. Affinity of W_q and cancellation of the two halves show that the two background-specific increments are equal. Taking h to be the constant- o_* profile and applying Lemma 32, then changing the finitely many branches one at a time, gives

$$W_q(C(P, g)) - W_q(C(P, o_*)) = \sum_y m_y [u(g(y)) - u(o_*)]. \quad (45)$$

The constant- o_* compound has exactly the information in the first-stage record. Lemma 31 therefore gives $W_q(C(P, o_*)) = u(o_*) + \lambda I_q(A; Y)$. Substitute this into (45) and use $\sum_y m_y = 1$. $\square$

B.4 Proof of the theorem

Proof of Theorem 1. Assume first Axioms (A1)–(A8). Let q have full support on a nontrivial action set, and let $K : A \rightarrow \Delta(O \times R)$ be arbitrary. Put $Y = O \times R$ and take as first stage

$$P_K(y \mid a) := K(y \mid a), \quad y = (o, r).$$

After $y = (o, r)$, pay o for sure and reveal nothing further. Call the resulting compound K^{seq} . From K , the record processor sends (o, r) to the sequential record $((o, r), *)$. In the reverse direction it sends $((o, r), *)$ to r ; inconsistent payoff-record pairs are unreached and may be completed arbitrarily. Both processors preserve the actual payoff coordinate. Hence Axiom (A6) gives $(q, K) \sim (q, K^{\text{seq}})$.

Apply Lemma 33 with $g(o, r) = o$. The two identities needed are

$$I_q(A; Y) = I_{qK}(A; O, R)$$

and

$$\sum_{o,r} p_{qK}(o, r) u(o) = \mathbb{E}_{qK}[u(O)],$$

respectively. Together with sequentialisation, they give

$$W_q(K) = \mathbb{E}_{qK}[u(O)] + \lambda I_{qK}(A; O, R). \quad (46)$$

If A is a singleton, mutual information is zero and the marked terminal law is that of the induced payoff lottery. Equation (37) gives (46) directly. If q is on the boundary, put $S = \text{supp } q$ and write (q^S, K^S) for the restriction. Then

$$(q, K) \sim (q^S, K^S), \quad V(q, K) = V(q^S, K^S). \quad (47)$$

The first identity is Lemma 28(b); the second follows directly from the finite sums defining expected utility and mutual information. Thus no representative W_q is assigned to a boundary fibre.

Restrict two arbitrary alternatives to their supports, apply (38) and (46) there, and use (47) to transport the comparison back. This gives

$$(q, K) \succeq (p, L) \iff V(q, K) \geq V(p, L), \quad (48)$$

with the same u and λ . Axiom (A5), first by duplication and then by deletion of irrelevant blocks, turns (48) into the within-channel representation and the finite-block moreover clause. The construction made u nonconstant and $\lambda > 0$. This proves the forward implication.

Conversely, suppose that a nonconstant u and a number $\lambda > 0$ represent the family by V . Axiom (A1) follows from the order on $\mathbb{R}$. Expected utility and mutual information are jointly continuous in (K, q) on each finite simplex; for the latter, use the entropy formula and $0 \log 0 = 0$. Hence the set of triples (K, q, p) for which $V(q, K) \geq V(p, K)$ is closed, which is Axiom (A2). A block-supported prior has a constant block label, which changes neither term of V ; this proves Axiom (A5) and reduces every pair comparison below to the corresponding two values.

A payoff-preserving record processor leaves the payoff marginal unchanged and weakly lowers $I(A; O, R)$ by data processing. This proves Axiom (A6). Under an action processor, equation (1) again leaves the payoff-record marginal unchanged, and the joint law forms the Markov chain

$$(O, R) - A - B.$$

Thus $I(B; O, R) \leq I(A; O, R)$, proving Axiom (A7). This argument uses the undivided joint-law equation and so also covers every completion on a null processed action.

For the binary compounds in Axiom (A8), the expected-utility and mutual-information chain rules give

$$\begin{aligned} V(q, P \triangleright (K, M)) - V(q, P \triangleright (L, M)) \\ = \left(\sum_a q(a) P(1 \mid a) \right) [V(q_1, K) - V(q_1, L)]. \end{aligned}$$

The first-stage information term and the continuation following record 2 cancel. Because record 1 has positive probability under q , the coefficient is positive, so the two differences have the same sign. This proves the weak biconditional in Axiom (A8) and its strict form.

Because u is nonconstant, choose o^+, o^- with $u(o^+) > u(o^-)$. In the two-action channel of Axiom (A3), its two pure lotteries have these values and zero information, so Axiom (A3) holds. For any $o \in O$, form the four-action channel described in Axiom (A4). Its two lotteries pay o surely and have information values $\log 2$ and zero. Since $\lambda > 0$, the required strict comparison holds—indeed, it holds at every o , although Axiom (A4) requires only one. The same block calculation used above gives the moreover clause with the original witnesses u and λ . $\square$

C Auxiliary results

This appendix collects proofs of the separability, uniqueness, sign, independence, and choice-implication results stated in the main text. None is used to prove Theorem 1; separating them keeps Appendices A and B focused on the representation argument.

C.1 Separability, uniqueness, and sign

For $q \in \Delta(A)$ and $K : A \rightarrow \Delta(O \times R)$, write $\pi_{or}^{qK}(a) := q(a)K(o, r \mid a)/p_{qK}(o, r)$ when $p_{qK}(o, r) > 0$ for the posterior induced by the visible pair, and $\mu_{qK} := \sum_{(o,r):p_{qK}(o,r)>0} p_{qK}(o, r) \delta_{\pi_{or}^{qK}}$ for its distribution.

Corollary 34 (No payoff–trace complementarity). *Suppose Axioms (A1)–(A8) hold. Fix A and $q \in \Delta(A)$. If channels K and L satisfy $\mathbb{E}_{qK}[u(O)] = \mathbb{E}_{qL}[u(O)]$ and $\mu_{qK} = \mu_{qL}$, then $(q, K) \sim (q, L)$.*

Proof. The result follows from Theorem 1 and $I_{qK}(A; O, R) = H(q) - \int H(\pi) d\mu_{qK}(\pi)$. $\square$

Proof of Proposition 2. Choose $o_* \in O$ with $u(o_*) = \underline{u}$. First note that the range of V is $[\underline{u}, \infty)$. The lower bound is immediate. For $q \in \Delta(A)$ and $t \in [0, 1]$, let a constant- o_* record channel reveal the realised action with probability t and otherwise report $*$:

$$E_t(o_*, a' \mid a) = t \mathbf{1}_{a'=a}, \quad E_t(o_*, * \mid a) = 1 - t.$$

Then $V(q, E_t) = \underline{u} + \lambda t H(q)$. At the uniform lottery on n actions these values fill $[\underline{u}, \underline{u} + \lambda \log n]$; the intervals are unbounded.

For part (i), let W be a common-scale representation and define $\varphi(v) := W(q, K)$ for any pair with $V(q, K) = v$. This is well defined: equality of the two V -values gives indifference in their two-block environment by Theorem 1, and the common-scale property gives equality of the two W -values. A strict inequality between V -values similarly gives a strict inequality between W -values. Thus φ is strictly increasing and $W = \varphi \circ V$. The converse is immediate from Theorem 1.

For part (ii), the expected-utility and mutual-information chain rules give, for every $P : A \rightarrow \Delta(\{1, \dots, k\})$ and every $K_1, \dots, K_k, L_1, \dots, L_k : A \rightarrow \Delta(O \times R)$, writing $m(i) := \sum_a q(a)P(i \mid a)$,

$$V(q, P \triangleright (K_1, \dots, K_k)) - V(q, P \triangleright (L_1, \dots, L_k)) = \sum_{i:m(i)>0} m(i) [V(q_i, K_i) - V(q_i, L_i)].$$

Hence every $\alpha V + \beta$, with $\alpha > 0$, is common-scale and branch-additive.

Conversely, let the branch-additive common-scale representation be $W = \varphi \circ V$ and define

$$\psi(x) := \varphi(\underline{u} + x) - \varphi(\underline{u}), \quad x \geq 0.$$

Fix $n \geq 2$ and let q be uniform on n actions. For $x \in [0, \lambda \log n]$, choose a constant- o_* erasure channel whose value is $\underline{u} + x$. In the first compound, use this channel after the first branch of a public coin with probabilities θ and $1 - \theta$, and use the uninformative constant- o_* channel after the second. In the comparison compound, use the uninformative channel after both branches. The two compound values are $\underline{u} + \theta x$ and $\underline{u}$, so (6) gives

$$\psi(\theta x) = \theta \psi(x) \quad (0 \leq \theta \leq 1).$$

Taking $x = \lambda \log n$ shows that ψ is linear on $[0, \lambda \log n]$. These intervals are nested and unbounded, so $\psi(x) = \alpha x$ on $[0, \infty)$ for one $\alpha > 0$. Therefore $\varphi(v) = \alpha v + \beta$ on the range of V .

Finally, any second trace-tempered index $\tilde{V}$ is common-scale, because expected utility and mutual information are invariant under block embedding, and is branch-additive by the same two chain rules. Applying part (ii) and then restricting to deterministic payoff outcomes and to constant-payoff record channels gives $\tilde{u} = \alpha u + \beta$ and $\tilde{\lambda} = \alpha \lambda$. $\square$

Proof of Corollary 3. Part (i) is Theorem 1. For part (ii), reverse the primitive relation in every environment. The reversed family satisfies Axioms (A1), (A2), (A5), and (A8); Axiom (A3) holds with its two witnessing outcomes interchanged. It also satisfies the forward versions of Axioms (A4), (A6), and (A7). Theorem 1 therefore represents the reversed family by $\mathbb{E}[\hat{u}(O)] + \hat{\lambda}I(A; O, R)$ with $\hat{\lambda} > 0$. Negating that index represents the original family, so it has the asserted form with $u = -\hat{u}$ and $\lambda = -\hat{\lambda} < 0$.

For part (iii), collapse the action alphabet of any pair (q, K) to a singleton. Axiom (A7⁰) makes the pair indifferent to the resulting one-action channel, whose output law is p_{qK} . Axiom (A6⁰) then erases its record without changing value. Thus every pair is indifferent to the one-action channel that delivers precisely its payoff marginal. Axiom (A5) makes these reductions coherent across blocks. On payoff lotteries, Axiom (A8) applied to an action-independent public coin gives mixture independence, Axiom (A2) gives continuity, and Axiom (A3) gives non-triviality. The mixture-space theorem therefore yields a nonconstant expected-utility index. Conversely, expected utility is unaffected by record or action processing and satisfies the three indifference clauses. The positive and negative formulae satisfy their respective clauses by data processing and the two chain rules. $\square$

C.2 Independence of the axioms

Proof of Proposition 4. Fix a nonconstant u and write $U(q, K) := \mathbb{E}_{qK}[u(O)]$, $Z := (O, R)$, and $I := I(A; Z)$. All scores below are extended to block and compound channels by the same formula.

Failure of Axiom (A1). Use the Pareto relation on the two coordinates (U, I) :

$$(q, K) \succeq (p, L) \iff U(q, K) \geq U(p, L) \text{ and } I(q, K) \geq I(p, L).$$

It is transitive but incomplete whenever one alternative has more payoff and the other more information. Its graph is closed. Both data-processing axioms are coordinatewise monotonic. Under a common-branch replacement, both the payoff and information differences are multiplied by the same positive branch probability, so Axiom (A8) holds in both directions. Block coherence holds. For Axiom (A3), choose the two outcomes so that $u(\sigma^+) > u(\sigma^-)$; for Axiom (A4), the two test lotteries have equal payoff value and information values $\log 2$ and zero. Thus both relevance axioms hold.

Failure of Axiom (A2). Order alternatives lexicographically by (U, I) , with payoff first. This is a weak order and satisfies all structural, processing, and continuation requirements. Axiom (A3) holds through the first coordinate, and Axiom (A4) through the second when the first is tied. It is not continuous even within one fixed channel. Let a three-action channel reveal the action fully and let only action 2 pay the high outcome. Put

$$p = (1/4, 1/2, 1/4), \quad q = (0.49, 0.50, 0.01), \quad q_n = (0.49 - \varepsilon_n, 0.50 + \varepsilon_n, 0.01),$$

where $\varepsilon_n \downarrow 0$. The lotteries p and q have the same expected utility but $H(q) < H(p)$, so $p \succ_K q$. Every q_n has strictly higher expected utility than p , so $q_n \succ_K p$. Thus the graph in Axiom (A2) is not closed.

Failure of Axiom (A3). Use $V := I$. All other axioms hold. In particular, the test in Axiom (A4) has information values $\log 2$ and zero. But the two pure lotteries in Axiom (A3) both have zero information, whatever their payoffs.

Failure of Axiom (A4). Use $V := U$. All other axioms hold. The outcomes in Axiom (A3) can be chosen to have different utility, but the two lotteries in Axiom (A4) always have the same payoff and hence are indifferent.

Failure of Axiom (A5). Treat outer block labels as syntactic action tags. For an action alphabet presented with at least three outer blocks, put $w(a^0) = 1$ and $w(a^i) = 0$ for $i \neq 0$; put

$w(a) = 0$ on every action alphabet with fewer than three outer blocks. For $\varepsilon > 0$, use within each environment

$$V_\varepsilon(q, K) := U(q, K) + I(q, K) + \varepsilon \sum_a q(a)w(a).$$

This defines a continuous weak order in every fixed environment. Every comparison in Axioms (A6), (A7), and (A8) is made in a two-block environment, where $w = 0$; those axioms therefore reduce to their valid $U + I$ comparisons. The fixed relevance tests can be chosen on untagged action sets and also reduce to $U + I$. But irrelevant-block coherence fails. Take three identical one-action, constant-payoff, silent channels. Their first two block-supported lotteries are indifferent in the canonical two-block comparison, while in the three-block environment the lottery supported on block 0 is strictly preferred to the lottery supported on block 1 by ε .

Failure of Axiom (A6). For $\varepsilon > 0$, use

$$V(q, K) := U(q, K) + I(A; Z) - \varepsilon H(Z | A). \quad (49)$$

This score is continuous and branch-additive, since both mutual information and conditional entropy obey the chain rule. Under action processing $A \rightarrow B$, the score difference between the original and processed alternatives is

$$[I(A; Z) - I(B; Z)] + \varepsilon [H(Z | B) - H(Z | A)] = (1 + \varepsilon)I(A; Z | B) \geq 0,$$

so (49) satisfies Axiom (A7). Axiom (A3) holds through U . In the Axiom (A4) test, the first lottery has $I = \log 2$ and $H(Z | A) = 0$, while the second has $I = 0$ and $H(Z | A) = \log 2$; hence the first is strictly preferred. Record data processing fails: with constant payoff and a fair record independent of A , the original value is $-\varepsilon \log 2$; deleting that noise gives value zero.

Failure of Axiom (A7). Let $G(q) := 1 - \sum_a q(a)^2$ be Gini entropy and define its posterior reduction

$$J_G(q, K) := G(q) - \int G(\pi) d\mu_{qK}(\pi), \quad V := U + J_G.$$

Concavity of G gives record data processing, and the potential-difference form gives the exact branch chain rule. Axiom (A3) holds through U . In the Axiom (A4) test, J_G is $1/2$ under the first lottery and zero under the second, so trace relevance holds. Action processing can nevertheless increase J_G . Take $q = (1/8, 1/8, 3/4)$, a binary record with

$$\Pr(0 | a_1) = \Pr(0 | a_2) = 0, \quad \Pr(0 | a_3) = 1/4,$$

and merge a_1, a_2 into one processed action. Direct calculation gives $J_G = 9/416$ before the merge and $J_G = 3/104$ afterwards. Hence Axiom (A7) fails.

Failure of Axiom (A8). Use $V := U + \min\{I, 1\}$, with information measured in bits. It is continuous, respects both data-processing axioms, and satisfies block coherence. Axiom (A3) holds through U , while the test in Axiom (A4) has information values one and zero. Let q be uniform on four actions. A first-stage record reveals which of two pairs contains the action. Hold the continuation following one record fixed and silent. Following the other record, keep payoffs constant and compare a silent continuation with one that reveals the action within its pair. The latter continuation is strictly preferred at the branch posterior, but the two compounds have respectively one and three-halves bits before capping and are therefore indifferent. The forward implication in Axiom (A8) fails. $\square$

C.3 Proofs of the choice implications

Several results stated only in summary form in the text are given here in full. Proofs appear in an order that front-loads shared machinery. Throughout this subsection, write $Z := O \times R$ and $K_a := K(\cdot | a) \in \Delta(Z)$ for the row of action a ; the distribution induced by q is p_{qK} , as in Section 3.

Proof of Proposition 5. (i) Under either lottery, each visible pair (o_0, r_j) , $j = 1, 2$, has probability $\frac{1}{2}$: under q' because each a_j produces (o_0, r_j) for sure, under q'' because each of a_3, a_4 produces either pair with equal probability. Hence $p_{q'K} = p_{q''K}$. On $\text{supp}(q')$ the rows are degenerate and distinct, so the visible pair reveals the action and $I_{q'K} = H(q') = \log 2$; on $\text{supp}(q'')$ the rows are identical, so the visible pair is independent of the action and $I_{q''K} = 0$. The payoff terms agree because the payoff marginals do, giving the displayed gap. (ii) All rows of K° are equal, so $I_{qK^\circ} = 0$ for every q ; since the payoff is constantly o_0 , the representation gives indifference. $\square$

Proof of Proposition 6. Expected utility is linear in q , while mutual information is concave in the input distribution of a fixed channel [16]. Continuity on the compact simplex gives existence. Using

$$I_{qK}(A; Z) = \sum_a q(a) D_{\text{KL}}(K_a \| p_{qK}),$$

its directional derivative in the mass on action a equals the displayed divergence up to a term common to every action. The result is the Kuhn–Tucker condition for a concave programme. At constant u it reduces to the classical capacity condition. $\square$

Proposition 35 (Optimal marginal and multiplicity). *Let $\lambda > 0$ and write $C^*(K) := \max_{q \in \Delta(A)} V(q, K)$. There is a unique $p^* \in \Delta(Z)$ such that $p_{q^*K} = p^*$ for every optimal lottery q^* . With*

$$\mathcal{A}^* := \{a \in A : \bar{u}(a) + \lambda D_{\text{KL}}(K_a \| p^*) = C^*(K)\},$$

the set of optimal lotteries is exactly

$$\{q \in \Delta(A) : p_{qK} = p^*, \text{ supp}(q) \subseteq \mathcal{A}^*\}.$$

The optimum is unique whenever the rows $\{K_a : a \in \mathcal{A}^\}$ are affinely independent. If actions are partitioned into classes with identical rows, $V(q, K)$ depends on q within each class only through the class's total mass. The optimal lotteries are exactly the lifts of the optimal lotteries of the collapsed channel, and each class's mass may be distributed arbitrarily among its members.*

Proof of Proposition 35. The entropy identity gives

$$V(q, K) = \lambda H(p_{qK}) + \sum_a q(a) [\bar{u}(a) - \lambda H(K_a)].$$

If q_0 and q_1 are optimal with $p_{q_0K} \neq p_{q_1K}$, strict concavity of Shannon entropy and linearity of $q \mapsto p_{qK}$ and of the remaining sum give $V(tq_0 + (1-t)q_1, K) > C^*(K)$ for $t \in (0, 1)$, a contradiction. Hence every optimum induces the same p^* . By Proposition 6, an optimum q^* satisfies the displayed equalities at some constant c ; averaging them under q^* gives $c = V(q^*, K) = C^*(K)$, so $\text{supp}(q^*) \subseteq \mathcal{A}^*$. Conversely, if $p_{qK} = p^*$ and $\text{supp}(q) \subseteq \mathcal{A}^*$, then

$$V(q, K) = \sum_a q(a) [\bar{u}(a) + \lambda D_{\text{KL}}(K_a \| p^*)] = C^*(K).$$

Affine independence makes the representation $p^* = \sum_a q(a) K_a$ with $\text{supp}(q) \subseteq \mathcal{A}^*$ unique. Identical rows share $\bar{u}(a)$ and $H(K_a)$, so both p_{qK} and the displayed expression for V are unchanged by redistribution within a class; collapsing the classes and lifting aggregate masses proves the last claim. $\square$

Proposition 36 (Support and pure choice). *Let $\lambda > 0$ and let $Z_K := \{z \in Z : K_a(z) > 0 \text{ for some } a \in A\}$.*

- (i) *Every optimal lottery q^* satisfies $p_{q^*K}(z) > 0$ for every $z \in Z_K$. Consequently, if every action a has a private consequence—some z_a with $K_a(z_a) > 0$ and $K_b(z_a) = 0$ for all $b \neq a$ —then every optimal lottery has full support.*

(ii) The pure lottery δ_a is optimal if and only if

$$\bar{u}(b) + \lambda D_{\text{KL}}(K_b \| K_a) \leq \bar{u}(a) \quad \text{for every } b \in A,$$

and uniquely optimal if these inequalities are strict for $b \neq a$. In particular, if some rival reaches a consequence outside the support of K_a , then δ_a is not optimal, however large its payoff advantage.

Proof of Proposition 36. Throughout, p^* denotes the common optimal marginal of Proposition 35.

(i) Suppose $p^*(z) = 0$ for some $z \in Z_K$ and choose a with $K_a(z) > 0$. Then $D_{\text{KL}}(K_a \| p^*) = +\infty$, which is incompatible with the finite on-support equality if some optimum uses a and with the off-support inequality of Proposition 6 if none does. Under the private-consequence condition, an optimum with $q^*(a) = 0$ would force $p^*(z_a) = 0$.

(ii) For $q = \delta_a$ the induced marginal is K_a and the on-support constant is $\bar{u}(a)$; the displayed system is the off-support condition of Proposition 6, necessary and sufficient by concavity. Under strict inequalities, $\mathcal{A}^* = \{a\}$ in Proposition 35, so every optimum is δ_a . $\square$

Proposition 37 (Optimal mixing). *Let $\lambda > 0$ and $A = \{a, b\}$ with $K_a \neq K_b$ and $\Delta := \bar{u}(a) - \bar{u}(b) \geq 0$. For $x \in [0, 1]$ let q_x put mass x on b , so that $p_{q_x K} = (1 - x)K_a + xK_b =: p_x$, and for $x \in (0, 1)$ define*

$$g(x) := D_{\text{KL}}(K_b \| p_x) - D_{\text{KL}}(K_a \| p_x).$$

(i) *g is continuous and strictly decreasing, with $g(0^+) = D_{\text{KL}}(K_b \| K_a)$ and $g(1^-) = -D_{\text{KL}}(K_a \| K_b)$ as extended-real limits. The pure lottery δ_a is optimal if and only if $\Delta \geq \lambda D_{\text{KL}}(K_b \| K_a)$. Below this threshold the optimum is unique and interior, $q^* = q_{x^*}$ with x^* the unique solution of $\Delta = \lambda g(x)$.*

(ii) *The optimal mass $x^*(\Delta, \lambda)$ on the inferior action is nonincreasing in Δ and nondecreasing in λ , strictly so on the mixing region; at $\Delta = 0$ it is the capacity-achieving input of the two-row channel, independently of λ .*

Proof. Write $f(x) := V(q_x, K) = \bar{u}(a) - x\Delta + \lambda I(x)$ with $I(x) = H(p_x) - (1 - x)H(K_a) - xH(K_b)$. Since $K_a \neq K_b$, $x \mapsto p_x$ is affine and injective, so strict concavity of H [16] makes f strictly concave on $[0, 1]$, and a unique maximiser exists. On $(0, 1)$ every $z \in \text{supp } K_a \cup \text{supp } K_b$ has $p_x(z) > 0$ and, using $\sum_z (K_b(z) - K_a(z)) = 0$,

$$I'(x) = H(K_a) - H(K_b) - \sum_z (K_b(z) - K_a(z)) \log p_x(z) = g(x),$$

so $f'(x) = -\Delta + \lambda g(x)$, with g continuous and strictly decreasing as the derivative of a strictly concave function. As $x \downarrow 0$, $p_x(z) \rightarrow K_a(z)$; if $\text{supp } K_b \subseteq \text{supp } K_a$ every summand converges and $g(0^+) = D_{\text{KL}}(K_b \| K_a)$, while if some $\bar{z}$ has $K_b(\bar{z}) > 0 = K_a(\bar{z})$ the summand $-K_b(\bar{z}) \log(xK_b(\bar{z}))$ diverges to $+\infty$, matching the extended-sense divergence. Exchanging the roles of a and b gives the limit at 1. By concavity and the mean value theorem, δ_a is optimal if and only if $\lim_{x \downarrow 0} f'(x) \leq 0$, i.e. $\Delta \geq \lambda D_{\text{KL}}(K_b \| K_a)$; this is the threshold stated in the text, and Proposition 36(ii) with two actions. If $\Delta < \lambda D_{\text{KL}}(K_b \| K_a)$, the maximiser is not $x = 0$ by the threshold and not $x = 1$ because $f'(1^-) = -\Delta - \lambda D_{\text{KL}}(K_a \| K_b) < 0$; it is therefore the unique root of $\Delta = \lambda g(x)$, which exists because g decreases strictly between its limits. This proves (i). The comparative statics of (ii) follow by inverting the strictly decreasing g at Δ/λ : raising Δ raises the target, raising λ lowers it; at $\Delta = 0$ the condition $g(x) = 0$ is the capacity condition of Proposition 6 at constant payoff. $\square$

Remark 38 (Overlapping traces and IIA). Let three actions have equal expected utility and rows

$$K_1 = (1, 0, 0), \quad K_2 = (0, 1, 0), \quad K_3 = (1/2, 0, 1/2).$$

With only the first two actions, the optimal lottery is $(1/2, 1/2)$. Adding the third gives

$$q^* = (1/3, 4/9, 2/9), \quad p_{q^*K} = (4/9, 4/9, 1/9).$$

Each active row has divergence $\log(9/4)$ from p_{q^*K} , so Proposition 6 verifies optimality. The ratio of the first two choice probabilities falls from 1 to $3/4$. Thus overlapping rows can generate the red-bus/blue-bus pattern even when expected utilities are equal.

Proposition 39 (Trace demand). *Fix K and write $e(q) := \mathbb{E}_{qK}[u(O)]$, $i(q) := I_{qK}(A; O, R)$,*

$$V^*(\lambda) := \max_{q \in \Delta(A)} \{e(q) + \lambda i(q)\}, \quad Q(\lambda) := \arg \max_{q \in \Delta(A)} \{e(q) + \lambda i(q)\}.$$

(i) V^* is convex on $(0, \infty)$, i is affine on the convex compact set $Q(\lambda)$, and

$$(V^*)'_-(\lambda) = \min_{q \in Q(\lambda)} i(q), \quad (V^*)'_+(\lambda) = \max_{q \in Q(\lambda)} i(q);$$

at every point of differentiability, $(V^*)'(\lambda) = i(q)$ for every $q \in Q(\lambda)$.

(ii) Let $0 < \lambda_1 < \lambda_2$ and $q_j \in Q(\lambda_j)$, writing $e_j := e(q_j)$, $i_j := i(q_j)$. Then $i_1 \leq i_2$, $e_1 \geq e_2$, and

$$\lambda_1(i_2 - i_1) \leq e_1 - e_2 \leq \lambda_2(i_2 - i_1).$$

Hence either $(e_1, i_1) = (e_2, i_2)$, or $i_1 < i_2$ and $e_1 > e_2$ and

$$\lambda_1 \leq \frac{e_1 - e_2}{i_2 - i_1} \leq \lambda_2.$$

Proof of Proposition 39. (i) V^* is a pointwise maximum of functions affine in λ , hence convex and, since $0 \leq i \leq \log |A|$, finite. On the convex compact set $Q(\lambda)$ the objective is constant and e is linear, so $i = (V^* - e)/\lambda$ is affine there. For $q \in Q(\lambda)$ and $\mu > 0$, $V^*(\mu) \geq V^*(\lambda) + (\mu - \lambda)i(q)$, so each such $i(q)$ is a subgradient. For the right derivative, choose $q_h \in Q(\lambda + h)$; then

$$\max_{q \in Q(\lambda)} i(q) \leq \frac{V^*(\lambda + h) - V^*(\lambda)}{h} \leq i(q_h),$$

the first inequality by the subgradient bound, the second by optimality of $Q(\lambda)$ at λ . By compactness and continuity, accumulation points of $(q_h)_{h \downarrow 0}$ lie in $Q(\lambda)$, so the right-hand side accumulates in $i(Q(\lambda))$ and the right derivative equals the maximum. The left derivative is symmetric, and a convex function on an interval is differentiable off a countable set.

(ii) Optimality at the two weights gives $e_1 + \lambda_1 i_1 \geq e_2 + \lambda_1 i_2$ and $e_2 + \lambda_2 i_2 \geq e_1 + \lambda_2 i_1$. Adding yields $(\lambda_2 - \lambda_1)(i_2 - i_1) \geq 0$, hence $i_2 \geq i_1$; the two inequalities rearrange to the displayed bracket, which gives $e_1 \geq e_2$ and, when the pairs differ, the final bound. $\square$

Corollary 40 (Extreme trace weights). *Let $e^0 := \max_q e(q)$, $i^0 := \max\{i(q) : e(q) = e^0\}$, $C := \max_q i(q)$, and $e^C := \max\{e(q) : i(q) = C\}$. If $\lambda_n \downarrow 0$ and $q_n \in Q(\lambda_n)$, every accumulation point of (q_n) maximises i among the maximisers of e ; if $\lambda_n \uparrow \infty$, every accumulation point maximises e among the maximisers of i . Moreover*

$$V^*(\lambda) = e^0 + \lambda i^0 + o(\lambda) \quad \text{as } \lambda \downarrow 0, \quad V^*(\lambda) = \lambda C + e^C + o(1) \quad \text{as } \lambda \uparrow \infty.$$

Proof of Corollary 40. For $\lambda_n \downarrow 0$, comparing q_n with an e -maximiser attaining i^0 gives

$$0 \leq e^0 - e(q_n) \leq \lambda_n(i(q_n) - i^0).$$

Boundedness of i forces $e(q_n) \rightarrow e^0$, and the same inequality gives $i(q_n) \geq i^0$; continuity delivers the accumulation claim. Dividing by λ_n ,

$$\frac{V^*(\lambda_n) - e^0}{\lambda_n} = i(q_n) - \frac{e^0 - e(q_n)}{\lambda_n},$$

where the last fraction lies in $[0, i(q_n) - i^0]$ and vanishes, proving the first expansion. For $\lambda_n \uparrow \infty$, comparing q_n with an i -maximiser attaining e^C gives $0 \leq \lambda_n(C - i(q_n)) \leq e(q_n) - e^C$, and the symmetric argument completes the proof. $\square$

Proof of Proposition 8. The processor leaves the payoff coordinate unchanged, so the expected-utility terms coincide and the difference is $\lambda[I_{qK}(A; O, R) - I_{qL}(A; O, R')]$. Write $Z := (O, R)$ and $Z' := (O, R')$. By construction $A \rightarrow Z \rightarrow Z'$ is a Markov chain, so expanding $I(A; Z, Z')$ in both orders gives

$$I(A; Z) - I(A; Z') = I(A; Z \mid Z') = I(A; R \mid O, R'),$$

the last equality because O is part of Z' . Conditional mutual information is nonnegative and vanishes exactly when A and (O, R) are conditionally independent given (O, R') [16]. This is the zero-price condition stated in the proposition. Under a garbling, it is also equivalent to equality of the posterior laws, $\mu_{qK} = \mu_{qL}$. Indeed, the Markov property gives $\Pr(A = \cdot \mid Z, Z') = \pi_Z^{qK}$ almost surely, while conditional independence gives $\Pr(A = \cdot \mid Z, Z') = \pi_{Z'}^{qL}$; the two random posteriors then coincide almost surely. Conversely, by the identity in the proof of Corollary 34, equality of the posterior laws gives equality of the information terms, and the payoff terms already agree. $\square$

Corollary 41 (The recorded coin). *Let $K, L : A \rightarrow \Delta(O \times R)$ and $t \in (0, 1)$. Let $tK \oplus (1 - t)L$ be the public mixture that draws an action-independent coin Y with $\Pr(Y = 0) = t$, applies K if $Y = 0$ and L otherwise, and includes the coin in the record; let $M_t := tK + (1 - t)L$ be the rowwise mixture, which hides it. Then, for every $q \in \Delta(A)$,*

- (i) $V(q, tK \oplus (1 - t)L) = tV(q, K) + (1 - t)V(q, L)$;
- (ii) $V(q, tK \oplus (1 - t)L) - V(q, M_t) = \lambda I(A; Y \mid O, R) \geq 0$, with equality if and only if A and Y are conditionally independent given (O, R) .

Proof of Corollary 41. Part (i) is the chain rule with an action-independent first stage: $I(A; Y) = 0$, and each reached branch reproduces the joint law of the corresponding constituent. Part (ii) follows from Proposition 8: deleting the coin coordinate is a deterministic payoff-preserving record processor carrying the public mixture to M_t . $\square$

Definition 42 (Garbling and more-capable orders). Let $K : A \rightarrow \Delta(O \times R)$ and $L : A \rightarrow \Delta(O \times R')$ have the same payoff kernels. Write $K \succeq_g L$ if $L = KT$ for some record processor T , and write $K \succeq_c L$ if $I_{qK}(A; O, R) \geq I_{qL}(A; O, R')$ for every $q \in \Delta(A)$; the latter is the *more-capable* order on the visible-consequence channels [32].

Proposition 43 (Uniform dominance is more-capable, not garbling). *Let K and L have identical payoff kernels.*

- (i) $K \succeq_c L$ if and only if $V(q, K) \geq V(q, L)$ for every $q \in \Delta(A)$.
- (ii) $K \succeq_g L \Rightarrow K \succeq_c L$, but the converse fails. There are channels for which $K \succeq_c L$ although $L \neq KT$ for every record processor T .

Proof of Proposition 43. (i) Because the payoff kernels coincide, $V(q, K) - V(q, L) = \lambda[I_{qK} - I_{qL}]$ with $\lambda > 0$. Uniform value dominance and $K \succeq_c L$ are therefore the same statement.

(ii) If $L = KT$, Proposition 8 gives $I_{qK} \geq I_{qL}$ at every q , and hence $K \succeq_g L \Rightarrow K \succeq_c L$. For failure of the converse, let $A = \{0, 1\}$, fix $o^* \in O$, and let both channels pay o^* surely. Let K carry the binary erasure record with erasure probability $\varepsilon = \frac{1}{2}$ and L the binary symmetric record with crossover $p = \frac{1}{5}$:

$$K(o^*, a | a) = 1 - \varepsilon, \quad K(o^*, e | a) = \varepsilon, \quad L(o^*, a | a) = 1 - p, \quad L(o^*, 1 - a | a) = p.$$

The payoff kernels agree row by row, so it suffices to compare informations, and the comparison is invariant to the base; compute in bits with h the binary entropy. For $q = (1 - t, t)$, grouping records gives

$$I_{qK} = (1 - \varepsilon) h(t), \quad I_{qL} = h(p * t) - h(p), \quad p * t := p(1 - t) + (1 - p)t.$$

By Mrs Gerber's Lemma [46], $v \mapsto h(p * h^{-1}(v))$ is convex on $[0, 1]$, with h^{-1} valued in $[0, \frac{1}{2}]$; since $h(p * t)$ is invariant under $t \mapsto 1 - t$, take $t \leq \frac{1}{2}$ and $v = h(t)$. The chord between $v = 0$ and $v = 1$ gives

$$h(p * t) \leq (1 - v) h(p) + v, \quad \text{hence} \quad I_{qL} \leq (1 - h(p)) h(t) \leq (1 - \varepsilon) h(t) = I_{qK},$$

the last step because $\varepsilon = \frac{1}{2} \leq h(\frac{1}{5})$. Thus $K \succeq_c L$. Now suppose $L = KT$ for a processor T , and write $t_r := T(0 | o^*, r)$ for $r \in \{0, 1, e\}$. Matching the probability of the processed record 0 in the two rows,

$$(1 - \varepsilon)t_0 + \varepsilon t_e = 1 - p, \quad (1 - \varepsilon)t_1 + \varepsilon t_e = p,$$

and subtracting gives $t_0 - t_1 = (1 - 2p)/(1 - \varepsilon) = \frac{6}{5} > 1$, impossible. Compositions of record processors are record processors, so no chain of garblings produces L from K either. $\square$

The next definition and proposition formalise the side-bet elicitation of §4.4. Let o^+ and o^- attain the maximum and minimum of u , write $\Delta u := u(o^+) - u(o^-) > 0$, and for $\beta \in [0, 1]$ let $w(\beta) := \beta u(o^+) + (1 - \beta)u(o^-)$.

Definition 44 (Side bet). For $K : A \rightarrow \Delta(O \times R)$, put $R^* := R \sqcup \{*\}$ and regard K as a channel on R^* by zero padding. For $\varepsilon \in [0, 1]$ and $\beta \in [0, 1]$, let $P_\varepsilon : A \rightarrow \Delta(\{1, 2\})$ be the public coin $P_\varepsilon(2 | a) = \varepsilon$, and let $B_\beta : A \rightarrow \Delta(O \times R^*)$ have the constant rows $B_\beta(o^+, * | a) = \beta$ and $B_\beta(o^-, * | a) = 1 - \beta$. Write

$$D_{\varepsilon, \beta} K := P_\varepsilon \triangleright (K, B_\beta),$$

as in (2).

Proposition 45 (Willingness to pay for trace). Fix $q \in \Delta(A)$, $p \in \Delta(B)$, and channels $K : A \rightarrow \Delta(O \times R)$ and $L : B \rightarrow \Delta(O \times S)$.

(i) For every $\varepsilon \in [0, 1]$ and $\beta \in [0, 1]$,

$$V(q, D_{\varepsilon, \beta} K) = (1 - \varepsilon) V(q, K) + \varepsilon w(\beta).$$

(ii) If $\varepsilon \in (0, 1)$ and $\beta, \beta' \in [0, 1]$ satisfy $(q, D_{\varepsilon, \beta} K) \sim (p, D_{\varepsilon, \beta'} L)$, then

$$\frac{\varepsilon}{1 - \varepsilon} [w(\beta) - w(\beta')] = V(p, L) - V(q, K).$$

When the material parts agree, $\mathbb{E}_{qK}[u(O)] = \mathbb{E}_{pL}[u(O)]$, the left-hand side equals $\lambda[I_{pL}(B; O, S) - I_{qK}(A; O, R)]$: information is priced linearly at rate λ .

(iii) Such a pair (β, β') exists if and only if $|V(p, L) - V(q, K)| \leq \frac{\varepsilon}{1 - \varepsilon} \Delta u$; a boundary response therefore brackets the value difference instead of measuring it.

Proof. (i) The material part of the compound is $(1 - \varepsilon)\mathbb{E}_{qK}[u(O)] + \varepsilon w(\beta)$. For the trace part, the coin is action-independent, so both branch posteriors equal q and the chain rule for the compound gives an information term of $(1 - \varepsilon)I_{qK}(A; O, R)$, the side-bet branch contributing zero. (ii) By the moreover clause of Theorem 1, the indifference holds if and only if $(1 - \varepsilon)V(q, K) + \varepsilon w(\beta) = (1 - \varepsilon)V(p, L) + \varepsilon w(\beta')$; rearrange, and substitute the representation when the material parts cancel. (iii) $w(\beta) - w(\beta')$ is continuous on $[0, 1]^2$ with range $[-\Delta u, \Delta u]$; apply the intermediate value theorem to the equation in (ii). $\square$

Lemma 46 (The information–payoff plane). *Write $\phi(q, K) := (I_{qK}(A; O, R), E_{qK})$, and let $\underline{u} := \min_o u(o)$ and $\bar{u} := \max_o u(o)$. The image of ϕ over all finite pairs (q, K) is exactly $[0, \infty) \times [\underline{u}, \bar{u}]$, and the indifference classes of the block-comparison relation are, in these coordinates, the nonempty intersections of the lines $E = c - \lambda I$ with that strip.*

Proof. Inclusion is immediate. Conversely, fix (t, w) in the strip, choose n with $\log n \geq t$, and on $A = \{1, \dots, n\}$ use continuity of entropy along a segment from a point mass to the uniform lottery to find q_t with $H(q_t) = t$. Let $K(o^+, a | a) = \beta$ and $K(o^-, a | a) = 1 - \beta$ with β chosen so that $w(\beta) = w$. The payoff lottery is action-independent with mean w , and the record reveals the action, so $\phi(q_t, K) = (t, w)$. The description of the indifference classes then reads the representation through ϕ , using the moreover clause of Theorem 1. $\square$

Proof of Proposition 9. The identity rearranges $E_{qK} + \lambda I_{qK} = E_{pL} + \lambda I_{pL}$. For uniqueness, let $\lambda' \neq \lambda$, put $t := \frac{1}{2} \min\{1, (\bar{u} - \underline{u})/\lambda\} > 0$, and use Lemma 46 to realise pairs with ϕ -coordinates $(t, \bar{u} - \lambda t)$ and $(0, \bar{u})$; they are indifferent, while $E + \lambda' I$ differs across them by $(\lambda - \lambda')t \neq 0$. $\square$

To compare agents, let both families be defined over the same outcome set O , on the common domain of all finite pairs, with mutual information in one fixed base. A channel B is *silent* if all its rows are equal; then the joint law under any p is a product, so $I_{pB} = 0$ and $V^i(p, B)$ is the expectation of u_i under the O -marginal of the common row. Since one-action channels realise every payoff lottery in $\Delta(O)$, silent pairs realise exactly the payoff lotteries.

Definition 47 (More trace-seeking). Family 2 is *more trace-seeking than* family 1 if

$$(q, K) \succeq^1 (p, B) \implies (q, K) \succeq^2 (p, B)$$

for all lotteries q, p , all channels K , and all silent channels B .

Proposition 48 (More trace-seeking). *Let $\{\succeq_K^1\}$ and $\{\succeq_K^2\}$ satisfy Axioms (A1)–(A8). Then family 2 is more trace-seeking than family 1 if and only if the representations can be chosen with a common utility and $\lambda_2 \geq \lambda_1$.*

Proof of Proposition 48. Throughout, block comparisons are read through the moreover clause of Theorem 1.

Suppose the representations are (u, λ_1) and (u, λ_2) with $\lambda_2 \geq \lambda_1$. For any (q, K) and silent (p, B) ,

$$V^2(q, K) - V^2(p, B) = [V^1(q, K) - V^1(p, B)] + (\lambda_2 - \lambda_1)I_{qK} \geq V^1(q, K) - V^1(p, B),$$

since $I_{qK} \geq 0$ and the silent pair's value is material under both. A weak (or strict) agent-1 preference therefore transfers to agent 2, and reading the display from right to left gives the mirrored clause $(p, B) \succeq^2 (q, K) \Rightarrow (p, B) \succeq^1 (q, K)$: the weak one-sided definition, its strict variant, and the two-sided variant coincide on this class.

Conversely, suppose $\{\succeq_K^2\}$ is more trace-seeking than $\{\succeq_K^1\}$, with representations (u_1, λ_1) and (u_2, λ_2) .

Step 1: common utility. Let $D := \{\mu - \nu : \mu, \nu \in \Delta(O)\}$ and $f_i(d) := \sum_o u_i(o)d(o)$; each $f_i \neq 0$ on D because u_i is nonconstant. Silent pairs realise every payoff lottery, and the

definition's left argument ranges over all pairs, so silent-against-silent instances are in its scope: $f_1(d) \geq 0 \Rightarrow f_2(d) \geq 0$ on D . If $f_1(d) = 0$, applying this to d and $-d$ gives $f_2(d) = 0$, so $\ker f_1 \subseteq \ker f_2$ and $f_2 = \alpha f_1$ for some $\alpha \in \mathbb{R}$. If $\alpha = 0$ then u_2 is constant, contradicting Axiom (A3) for family 2. (The case is not idle: an index with constant u_2 , which violates only that axiom, is indifferent across all silent pairs and is more trace-seeking than every family whatsoever.) If $\alpha < 0$, pick o^+, o^- with $u_1(o^+) > u_1(o^-)$: the silent instance $\delta_{o^+} \succ^1 \delta_{o^-}$ forces $\delta_{o^+} \succeq^2 \delta_{o^-}$, while $\alpha < 0$ gives the strict reverse. Hence $\alpha > 0$ and $u_2 = \alpha u_1 + \beta$. Since $((u_2 - \beta)/\alpha, \lambda_2/\alpha)$ again represents family 2, choose $u_2 = u_1 =: u$, replacing λ_2 by $\lambda_2/\alpha > 0$. The choice is innocuous: any two common-utility normalisations rescale both weights by one positive factor, because u nonconstant forces the factors to agree, so the order of the weights does not depend on it.

Step 2: weight ordering. Write $\bar{u} := \max_o u(o)$ and $\underline{u} := \min_o u(o)$; $\bar{u} > \underline{u}$ by Axiom (A3). Suppose $\lambda_2 < \lambda_1$ and fix $\lambda' \in (\lambda_2, \lambda_1)$. Choose $t \in (0, (\bar{u} - \underline{u})/\lambda')$ and $w := \underline{u} + \lambda' t \in (\underline{u}, \bar{u})$. By Lemma 46 there is a pair (q, K) with $I_{qK} = t$ and $E_{qK} = \underline{u}$; its payoff lottery is action-independent, so after Step 1 both families assign it the same coordinates. Choose a silent (p, B) whose common-row O -marginal pays o^+ with probability $(w - \underline{u})/(\bar{u} - \underline{u})$ and o^- otherwise, so $V^1(p, B) = V^2(p, B) = w$. Then $V^1(q, K) = \underline{u} + \lambda_1 t > w$, so $(q, K) \succ^1 (p, B)$, and the definition gives $(q, K) \succeq^2 (p, B)$; but $V^2(q, K) = \underline{u} + \lambda_2 t < w$, a contradiction. Hence $\lambda_2 \geq \lambda_1$. $\square$

Remark 49. The proof uses $\lambda_i > 0$ only in the choice $E_{qK} = \underline{u}$; taking $E_{qK} = \bar{u}$ when $\lambda' < 0$ runs both directions verbatim. The equivalence therefore holds across all three orientations of Corollary 3: more trace-seeking and less trace-averse are one comparative.

Remark 50 (Eliciting λ). Normalise $u(o^+) = 1$ and $u(o^-) = 0$. All comparisons below are block comparisons, hence within the primitive domain, and no information quantity is estimated: the information gap is an accounting identity of the design. *Step 1.* For each $o \in O$, elicit β_o with $(\delta_*, K_o) \sim (\delta_*, B_{\beta_o})$, where K_o is the degenerate one-action channel used in Lemma 12; one-action channels carry zero information, so $u(o) = \beta_o$. *Step 2.* Fix $o \in O$, $n \geq 2$, the uniform q on $\{1, \dots, n\}$, a side-bet probability $\varepsilon \in (0, 1)$, and the constant-payoff channels K_o^{id} and K_o^0 used in Lemma 12. Elicit β with

$$(q, D_{\varepsilon, \beta} K_o^0) \sim (q, D_{\varepsilon, 0} K_o^{\text{id}}); \quad \text{then} \quad \lambda = \frac{\varepsilon}{1 - \varepsilon} \cdot \frac{\beta}{\log n},$$

by Proposition 45(i) applied to $V(q, K_o^{\text{id}}) = u(o) + \lambda \log n$ and $V(q, K_o^0) = u(o)$. Such a β exists if and only if $\lambda \log n \leq \varepsilon/(1 - \varepsilon)$; a subject still strictly preferring the right block at $\beta = 1$ reveals $\lambda > \varepsilon/((1 - \varepsilon) \log n)$, and ε is raised. *Controls.* Orientation: Theorem 1 implies $(q, K_o^{\text{id}}) \succ (q, K_o^0)$ at every payoff, although Axiom (A4) assumes the comparison at only one. Garbling: under the total record garbling of K_o^{id} , both blocks carry zero information and indifference must obtain; a surviving strict preference indicates that an uncontrolled feature of the display still traces the action. Transport: repeating Step 2 at another n , prior, or payoff level must return the same λ , by Proposition 9; an estimate that declines in $\log n$ is the signature of the capped-information criterion.

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